Main Category	Category	Terms
Disease	OMIM_DISEASE	Dyskeratosis congenita, digenic
Disease	UP_KW_DISEASE	Disease variant
Disease	UP_KW_DISEASE	Dyskeratosis congenita
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Nucleotide biosynthesis
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Membrane
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Mitochondrion
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Mitochondrion inner membrane
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Methyltransferase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Transferase
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_KW_PTM	Isopeptide bond
Functional Annotations	UP_SEQ_FEATURE	ACT_SITE:Nucleophile
Functional Annotations	UP_SEQ_FEATURE	BINDING:in other chain
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic and acidic residues
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cter in SUMO2)
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Thymidylate synthase/dCMP hydroxymethylase
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Gene Ontology	GOTERM_BP_DIRECT	dTMP biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	dTTP biosynthetic process

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	nucleotide biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of translation
Gene Ontology	GOTERM_BP_DIRECT	methylation
Gene Ontology	GOTERM_BP_DIRECT	tetrahydrofolate interconversion
Gene Ontology	GOTERM_BP_DIRECT	tetrahydrofolate metabolic process
Gene Ontology	GOTERM_BP_DIRECT	DNA biosynthetic process
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	mitochondrion
Gene Ontology	GOTERM_CC_DIRECT	mitochondrial inner membrane
Gene Ontology	GOTERM_CC_DIRECT	mitochondrial matrix
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	membrane
Gene Ontology	GOTERM_MF_DIRECT	mRNA regulatory element binding translation repressor activity
Gene Ontology	GOTERM_MF_DIRECT	thymidylate synthase activity
Gene Ontology	GOTERM_MF_DIRECT	folic acid binding
Gene Ontology	GOTERM_MF_DIRECT	methyltransferase activity
Gene Ontology	GOTERM_MF_DIRECT	transferase activity
Gene Ontology	GOTERM_MF_DIRECT	transferase activity, transferring one-carbon groups
Gene Ontology	GOTERM_MF_DIRECT	sequence-specific mRNA binding
General Annotations	CHROMOSOME	18

Main Category	Category	Terms
General Annotations	CYTOGENETIC_LOCATION	18p11.32
General Annotations	ENTREZ_GENE_SUMMARY	Thymidylate synthase catalyzes the methylation of deoxyuridylate to deoxythymidylate using, 10-methylenetetrahydrofolate (methylene-THF) as a cofactor. This function maintains the dTMP (thymidine-5-prime monophosphate) pool critical for DNA replication and repair. The enzyme has been of interest as a target for cancer chemotherapeutic agents. It is considered to be the primary site of action for 5-fluorouracil, 5-fluoro-2-prime-deoxyuridine, and some folate analogs. Expression of this gene and that of a naturally occurring antisense transcript, mitochondrial enolase superfamily member 1 (GeneID:55556), vary inversely when cell-growth progresses from late-log to plateau phase. Polymorphisms in this gene may be associated with etiology of neoplasia, including breast cancer, and response to chemotherapy. [provided by RefSeq, Aug 2017]
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing; Named isoforms=3; Name=1; IsoId=P04818-1; Sequence=Displayed; Name=2; Synonyms=delta4; IsoId=P04818-2; Sequence=VSP_047746; Name=3; Synonyms=delta2+3; IsoId=P04818-3; Sequence=VSP_047745;
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=dUMP + (6R)-5,10-methylene-5,6,7,8-tetrahydrofolate = 7,8- dihydrofolate + dTMP; Xref=Rhea:RHEA:12104, ChEBI:CHEBI:15636, ChEBI:CHEBI:57451, ChEBI:CHEBI:63528, ChEBI:CHEBI:246422; EC=2.1.1.45; Evidence={ECO:0000256|ARBA:ARBA00050752}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:12105; Evidence={ECO:0000256|ARBA:ARBA00050752};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=dUMP + (6R)-5,10-methylene-5,6,7,8-tetrahydrofolate = 7,8- dihydrofolate + dTMP; Xref=Rhea:RHEA:12104, ChEBI:CHEBI:15636, ChEBI:CHEBI:57451, ChEBI:CHEBI:63528, ChEBI:CHEBI:246422; EC=2.1.1.45; Evidence={ECO:0000269|PubMed:11278511}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:12105; Evidence={ECO:0000305|PubMed:11278511};
General Annotations	UP_COMMENT_DISEASE	Dyskeratosis congenita, digenic (DKCD) [MIM:620040]: A form of dyskeratosis congenita, a rare multisystem disorder caused by defective telomere maintenance. It is characterized by progressive bone marrow failure, and the clinical triad of reticulated skin hyperpigmentation, nail dystrophy, and mucosal leukoplakia. Common but variable features include premature graying, aplastic anemia, low platelets, osteoporosis, pulmonary fibrosis, and liver fibrosis among others. Early mortality is often associated with bone marrow failure, infections, fatal pulmonary complications, or malignancy. DKCD transmission pattern is consistent with digenic inheritance.

Main Category	Category	Terms
		{ECO:0000269|PubMed:35931051}. Note=The disease is caused by variants affecting distinct genetic loci, including the gene represented in this entry. TYMS germline variants in the presence of a common ENOSF1 haplotype (defined by rs699517, rs2790 and rs1512643) result in severe thymidylate synthase deficiency and disease. The pathogenic mechanism involves increased expression of ENOSF1 relative to TYMS, and post- transcriptional inhibition of TYMS translation through ENOSF1-TYMS RNA- RNA interactions. {ECO:0000269|PubMed:35931051}.
General Annotations	UP_COMMENT_FUNCTION	Catalyzes the reductive methylation of 2'-deoxyuridine 5'- monophosphate (dUMP) to thymidine 5'- monophosphate (dTMP), using the cosubstrate, 5,10- methylenetetrahydrofolate (CH2H4folate) as a 1- carbon donor and reductant and contributes to the de novo mitochondrial thymidylate biosynthesis pathway. {ECO:0000256|ARBA:ARBA00056634}.
General Annotations	UP_COMMENT_FUNCTION	Catalyzes the reductive methylation of 2'-deoxyuridine 5'- monophosphate (dUMP) to thymidine 5'- monophosphate (dTMP), using the cosubstrate, 5,10- methylenetetrahydrofolate (CH2H4folate) as a 1- carbon donor and reductant and contributes to the de novo mitochondrial thymidylate biosynthesis pathway. {ECO:0000269|PubMed:11278511, ECO:0000269| PubMed:21876188}.
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform 2]: Expressed both in normal and cancerous tissues. {ECO:0000305}.
General Annotations	UP_COMMENT_MISCELLANEOUS	[Isoform 3]: Expressed only in cancerous tissues. {ECO:0000305}.
General Annotations	UP_COMMENT_PATHWAY	Pyrimidine metabolism; dTTP biosynthesis. {ECO:0000256|ARBA:ARBA00004992}.
General Annotations	UP_COMMENT_PATHWAY	Pyrimidine metabolism; dTTP biosynthesis. {ECO:0000305|PubMed:11278511}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the thymidylate synthase family. {ECO:0000256|ARBA:ARBA00009972}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the thymidylate synthase family. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000269|PubMed:21876188}. Cytoplasm {ECO:0000269|PubMed:21876188}. Mitochondrion {ECO:0000269|PubMed:21876188}. Mitochondrion matrix {ECO:0000269|PubMed:21876188}. Mitochondrion inner membrane {ECO:0000269| PubMed:21876188}.
General Annotations	UP_COMMENT_SUBUNIT	Homodimer. {ECO:0000250|UniProtKB:P45352}.
Identifiers	ENTREZ_GENE_ID	7298
Identifiers	ENSEMBL_GENE_ID	ENSG00000176890
Identifiers	UNIPROT_ID	TYSY_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	Q53Y97_HUMAN
Identifiers	UNIPROT_ID	Q53FB7_HUMAN
Identifiers	UNIPROT_ID	A8K9A5_HUMAN
Interactions	DRUGBANK	Raltitrexed
Interactions	DRUGBANK	Floxuridine
Interactions	DRUGBANK	Trifluridine
Interactions	DRUGBANK	Gemcitabine
Interactions	DRUGBANK	Fluorouracil
Interactions	DRUGBANK	Methotrexate
Interactions	DRUGBANK	Pemetrexed
Interactions	DRUGBANK	Capecitabine
Interactions	DRUGBANK	Thymidine monophosphate
Interactions	DRUGBANK	10-Propargyl-5,8-Dideazafolic Acid
Interactions	DRUGBANK	Deoxyuridine monophosphate
Interactions	DRUGBANK	S,S-(2-Hydroxyethyl)Thiocysteine
Interactions	DRUGBANK	Thymectacin
Interactions	DRUGBANK	ANX-510
Interactions	DRUGBANK	OSI-7904L
Interactions	DRUGBANK	Pralatrexate
Interactions	DRUGBANK	2,4-Diamino-5-phenyl-6-ethylpyrimidine
Interactions	DRUGBANK	N-[2-Chloro-5-(trifluoromethyl)phenyl]imidodicarbonimidic diamide
Interactions	DRUGBANK	N-(3,5-dimethoxyphenyl)imidodicarbonimidic diamide
Interactions	DRUGBANK	6,6-DIMETHYL-1-[3-(2,4,5-TRICHLOROPHENOXY)PROPOXY]-1,6-DIHYDRO-1,3,5-TRIAZINE-2,4-DIAMINE
Interactions	DRUGBANK	Tegafur
Interactions	DRUGBANK	Tegafur-uracil
Interactions	PUBCHEM	N*6*-(2-Ethyl-pyridin-4-ylmethyl)-N*6*-methyl-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	N*6*-Methyl-N*6*-(2-phenylsulfanyl-pyridin-4-ylmethyl)-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	5,N*6*-Dimethyl-N*6*-pyridin-4-ylmethyl-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	N*6*-Methyl-N*6*-(2-methyl-pyridin-4-ylmethyl)-

Main Category	Category	Terms
		benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	5,N*6*-Dimethyl-N*6*-pyridazin-4-ylmethyl-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	2-Amino-6-methyl-5-[(4-nitrophenyl)sulfanyl]thieno[2,3-d]pyrimidin-4(3H)-one
Interactions	PUBCHEM	(2S)-2-{[4-({2,4-diamino-5-methyl-7H-pyrrolo[2,3-d]pyrimidin-6-yl}sulfanyl)phenyl]formamido} pentanedioic acid
Interactions	PUBCHEM	N*6*-Methyl-N*6*-(2-trifluoromethyl-pyridin-4-ylmethyl)-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	Lhkfzsdumipymf-uhfffaoysa-
Interactions	PUBCHEM	N-({4-[(2-amino-6-methyl-4-oxo-3,4-dihydrothieno[2,3-d]pyrimidin-5-yl)sulfanyl]phenyl}carbonyl)-L-glutamic acid
Interactions	PUBCHEM	5,N*6*-Dimethyl-N*6*-(2-methyl-pyridin-4-ylmethyl)-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	(4-{[(2-Amino-benzo[cd]indol-6-yl)-methyl-amino]-methyl}-pyridin-2-yl)-methanol
Interactions	PUBCHEM	(S)-2-{4-[2-(2-Amino-4-ethyl-6-methyl-7H-pyrrolo[2,3-d]pyrimidin-5-yl)-ethyl]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[2-amino-5-[[1-[(2R,4S,5R)-4-hydroxy-5-(phosphonooxymethyl)oxolan-2-yl]-2,4-dioxopyrimidin-5-yl]methyl]-4-oxo-4a,6,7,8-tetrahydropyrido[3,2-d]pyrimidin-6-yl]methylamino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	Methotrexate
Interactions	PUBCHEM	N(6)-(4-(piperazinylsulfonyl)benzyl)-6-aminobenz(cd)indol-2(1H)-one
Interactions	PUBCHEM	2-amino-6-ethyl-5-[(4-fluorophenyl)sulfanyl]-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	Raltitrexed
Interactions	PUBCHEM	Nolatrexed
Interactions	PUBCHEM	N-(4-(((3,4-Dihydro-2-methyl-4-oxo-6-quinazolinyl)methyl)-2-propyn-1-ylamino)benzoyl)-L-glutamic acid
Interactions	PUBCHEM	CB 3717
Interactions	PUBCHEM	Osi-7904
Interactions	PUBCHEM	(2S)-2-{[4-({2-amino-6-methyl-4-oxo-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-5-yl}sulfanyl)phenyl]formamido} pentanedioic acid
Interactions	PUBCHEM	(S)-2-{4-[(3-Amino-1-oxo-1,2-dihydro-benzo[f]quinazolin-9-ylmethyl)-amino]-benzoylamino}-pentanedioic acid

Main Category	Category	Terms
Interactions	PUBCHEM	6-[(4-benzoyl-N-prop-2-ynylanilino)methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	6-[(3-iodo-N-prop-2-ynylanilino)methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2,6-Dimethyl-5-(2-methyl-pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	Pemetrexed
Interactions	PUBCHEM	2-Methyl-6-[(phenyl-prop-2-ynyl-amino)-methyl]-3H-quinazolin-4-one
Interactions	PUBCHEM	6-Hydroxy-2-methyl-5-(pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	Pemetrexed Disodium
Interactions	PUBCHEM	2-methyl-6-[(3-methyl-N-prop-2-ynylanilino)methyl]-3H-quinazolin-4-one
Interactions	PUBCHEM	3-Amino-9-bromobenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	N-{4-[(2-Amino-4-Hydroxy-7h-Pyrrolo[2,3-D]pyrimidin-5-Yl)methyl]benzoyl}-L-Glutamic Acid
Interactions	PUBCHEM	2-amino-5-[(3-bromophenyl)sulfanyl]-6-ethyl-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-[(4-bromophenyl)sulfanyl]-6-ethyl-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	N*2*,N*6*-Dimethyl-N*6*-pyridazin-4-ylmethyl-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	(S)-2-({5-[(3-Methyl-1-oxo-1,2-dihydro-benzo[f]quinazolin-9-ylmethyl)-amino]-thiophene-2-carbonyl}-amino)-pentanedioic acid
Interactions	PUBCHEM	(S)-2-{2-Fluoro-4-[(3-methyl-1-oxo-1,2-dihydro-benzo[f]quinazolin-9-ylmethyl)-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	4-[[(7-Bromo-3,4-dihydro-2-methyl-4-oxo-6-quinazolinyl)methyl]-2-propyn-1-ylamino]-N-(4-pyridinylmethyl)benzamide
Interactions	PUBCHEM	(2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)sulfanyl]-2-fluorobenzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	5-(2-Dimethylamino-pyridin-4-ylsulfanyl)-2,6-dimethyl-3H-quinazolin-4-one
Interactions	PUBCHEM	Methopterin
Interactions	PUBCHEM	10-Propargyl-5,8-dideazafolic acid
Interactions	PUBCHEM	4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzonitrile
Interactions	PUBCHEM	6-[[3-(1-hydroxyethyl)-N-prop-2-ynylanilino]methyl]-2-

Main Category	Category	Terms
		methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	N*2*,N*6*-Dimethyl-N*6*-pyridin-4-ylmethyl-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	5-(2-Methoxy-pyridin-4-ylsulfanyl)-2,6-dimethyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2,6-Dimethyl-5-(pyridazin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	2,6-Dimethyl-5-(pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	6-{[(4-Benzenesulfonyl-phenyl)-prop-2-ynyl-amino]-methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-{4-[(2-Amino-4-hydroxy-quinazolin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	4-(2,6-Dimethyl-4-oxo-3,4-dihydro-quinazolin-5-ylsulfanyl)-benzoic acid
Interactions	PUBCHEM	2-amino-6-ethyl-5-(naphthalen-2-ylsulfanyl)-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(4-chloro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	6-[(3-ethynyl-N-prop-2-ynylanilino)methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-chloro-4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzonitrile
Interactions	PUBCHEM	2-Amino-5-(3,4-dimethoxy-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]-2-(trifluoromethyl)benzonitrile
Interactions	PUBCHEM	5,N*2*,N*6*-Trimethyl-N*6*-pyridin-4-ylmethyl-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	5,N*2*,N*6*-Trimethyl-N*6*-pyridazin-4-ylmethyl-benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	2-Amino-5-(2-dimethylamino-pyridin-4-ylsulfanyl)-6-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2,6-Dimethyl-5-(2-trifluoromethyl-pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	2-methyl-6-[[4-(4-methylpiperazin-1-yl)sulfonyl-N-prop-2-ynylanilino]methyl]-3H-quinazolin-4-one
Interactions	PUBCHEM	(S)-2-{4-[Methyl-(3-methyl-1-oxo-1,2-dihydro-benzo[f]quinazolin-9-ylmethyl)-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	2,6-Dimethyl-5-(pyridin-3-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	2,6-Dimethyl-5-(4-nitro-phenylsulfanyl)-3H-quinazolin-4-one

Main Category	Category	Terms
Interactions	PUBCHEM	(S)-2-{6-[(3-Methyl-1-oxo-1,2-dihydro-benzo[f]quinazolin-9-ylmethyl)-amino]-3-oxo-3H-benzo[d]isothiazol-2-yl}-pentanedioic acid
Interactions	PUBCHEM	methyl 3-[(1S)-1-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-2-methoxy-2-oxoethyl]benzoate
Interactions	PUBCHEM	2-amino-5-[(3,4-dichlorophenyl)sulfanyl]-6-ethyl-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzenesulfonamide
Interactions	PUBCHEM	4-[(7-bromo-2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]-N-(pyridin-2-ylmethyl)benzamide
Interactions	PUBCHEM	2-amino-6-ethyl-5-(naphthalen-1-ylsulfanyl)-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(4-bromo-3-nitro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	(2S)-2-{[4-({2-amino-6-ethyl-4-oxo-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-5-yl}sulfanyl)phenyl]formamido} pentanedioic acid
Interactions	PUBCHEM	methyl (2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-2-phenylacetate
Interactions	PUBCHEM	6-({[4-(2-Chloro-benzenesulfonyl)-phenyl]-prop-2-ynyl-amino}-methyl)-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	(S)-2-{5-[Methyl-(3-methyl-1-oxo-1,2-dihydro-benzo[f]quinazolin-9-ylmethyl)-amino]-1-oxo-1,3-dihydro-isoindol-2-yl}-pentanedioic acid
Interactions	PUBCHEM	2-amino-5-[(3-chlorophenyl)sulfanyl]-6-ethyl-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Methyl-5-(pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Amino-6-methyl-5-(pyridazin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	6-[[3-(hydroxymethyl)-N-prop-2-ynylanilino]methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	Xuefyxluosrufa-uhfffaoysa-
Interactions	PUBCHEM	2-methyl-6-[(4-piperazin-1-ylsulfonyl-N-prop-2-ynylanilino)methyl]-3H-quinazolin-4-one
Interactions	PUBCHEM	6-{[(4-Benzenesulfonyl-3-trifluoromethyl-phenyl)-prop-2-ynyl-amino]-methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Amino-5-(4-bromo-3-chloro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-methyl-6-[[4-nitro-N-prop-2-ynyl-3-(trifluoromethyl)anilino]methyl]-3H-quinazolin-4-one
Interactions	PUBCHEM	6-{[(4-Benzenesulfinyl-phenyl)-prop-2-ynyl-amino]-

Main Category	Category	Terms
		methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]-2-(trifluoromethyl)benzenesulfonamide
Interactions	PUBCHEM	2-amino-6-methyl-5-[(4-nitrophenyl)sulfanyl]-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	Rxxxgxvneqbqjg-uhfffaoysa-
Interactions	PUBCHEM	6-[(3-bromo-N-prop-2-ynylanilino)methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	6-[[4-(benzenesulfonyl)anilino]methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Amino-6-methyl-5-(pyridin-4-ylsulfanyl)-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(4-fluoro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	6-Ethyl-2-methyl-5-(pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Amino-6-chloro-5-(pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Amino-6-methyl-5-(2-trifluoromethyl-pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	5-(4-Benzenesulfonyl-phenylsulfanyl)-2,6-dimethyl-3H-quinazolin-4-one
Interactions	PUBCHEM	Qqfnuehbqptgkj-uhfffaoysa-
Interactions	PUBCHEM	2-[[4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]-2-(trifluoromethyl)phenyl]sulfonylamino]acetic acid
Interactions	PUBCHEM	6-Methoxy-2-methyl-5-(pyridin-4-ylsulfanyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	(2S)-2-[[5-[methyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	3-Aminobenzo[f]quinazolin-1-ol
Interactions	PUBCHEM	CB30865
Interactions	PUBCHEM	6-[(3-ethyl-N-prop-2-ynylanilino)methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Methyl-6-{[prop-2-ynyl-(3-trifluoromethyl-phenyl)-amino]-methyl}-3H-quinazolin-4-one
Interactions	PUBCHEM	2-amino-6-ethyl-5-[(4-nitrophenyl)sulfanyl]-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-methyl-6-[(4-nitro-N-prop-2-ynylanilino)methyl]-3H-quinazolin-4-one

Main Category	Category	Terms
Interactions	PUBCHEM	2-amino-5-[(3,4-dichlorophenyl)sulfanyl]-6-methyl-3H,4H,7H-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	6-{[(4-Benzenesulfinyl-3-chloro-phenyl)-prop-2-ynyl-amino]-methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-{4-[(4-Hydroxy-quinazolin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	2-{2-Fluoro-4-[(4-hydroxy-2-methyl-quinazolin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	2-{4-[(2-Amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-ylmethyl)-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	6-[[4-(benzenesulfonyl)-N-ethylanilino]methyl]-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	N-{4-[(2-Amino-6-Methyl-4-Oxo-4,7-Dihydro-3h-Pyrrolo[2,3-D]pyrimidin-5-Yl)sulfanyl]-2-Chlorobenzoyl}-L-Glutamic Acid
Interactions	PUBCHEM	(S)-2-{4-[(3-Methyl-1-oxo-1,2-dihydro-benzo[f]quinazolin-9-ylmethyl)-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	AG 85
Interactions	PUBCHEM	2-Amino-5-(Phenylsulfanyl)-3,9-Dihydro-4h-Pyrimido[4,5-B]indol-4-One
Interactions	PUBCHEM	2-Amino-5-(1-Naphthylsulfanyl)-3,9-Dihydro-4h-Pyrimido[4,5-B]indol-4-One
Interactions	PUBCHEM	2-{4-[(4-Oxo-2-trifluoromethyl-3,4-dihydro-quinazolin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	CB 3705
Interactions	PUBCHEM	3-amino-9-ethynyl benzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	N(10)-propargylfolic acid
Interactions	PUBCHEM	2-Desamino-2-methyl-5,8-dideazaisofolic acid
Interactions	PUBCHEM	CB 3988
Interactions	PUBCHEM	2-[4-(3-Methyl-1-oxo-1,2-dihydro-benzo[f]quinazoline-7-sulfonylamino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-Desamino-2-trifluoromethyl-N(10)-propargyl-5,8-dideazafolic acid
Interactions	PUBCHEM	(S)-2-(4-(((2-ethyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((2-(hydroxymethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid

Main Category	Category	Terms
Interactions	PUBCHEM	(S)-2-(4-(((2-methoxy-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((2-fluoroethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)-2-fluorobenzamido)pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[2-fluoro-4-[methyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(methyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(ethyl((2-(fluoromethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[4-[ethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-2-nitrobenzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((4-oxo-2-((pyrimidin-2-ylthio)methyl)-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-hydroxy-4-(((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	2-{4-[(2-Methyl-4-oxo-3,4-dihydro-quinazolin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	(2S)-2-{[4-({2-amino-6-methyl-4-oxo-3H,4H,5H-pyrrolo[3,2-d]pyrimidin-5-yl}methyl)phenyl]formamido} pentanedioic acid
Interactions	PUBCHEM	2-amino-6-methyl-5-(4-pyridylmethyl)-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-[(4-fluorophenyl)methyl]-6-methyl-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-[(4-chlorophenyl)methyl]-6-methyl-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-[(4-bromophenyl)methyl]-6-methyl-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-6-methyl-5-[(4-nitrophenyl)methyl]-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-[(4-methoxyphenyl)methyl]-6-methyl-3H-pyrrolo[3,2-d]pyrimidin-4-one

Main Category	Category	Terms
Interactions	PUBCHEM	2-amino-6-methyl-5-[[4-(trifluoromethoxy)phenyl]methyl]-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-[(3,5-dibromophenyl)methyl]-6-methyl-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-[(3,4-dichlorophenyl)methyl]-6-methyl-3H-pyrrolo[3,2-d]pyrimidin-4-one
Interactions	PUBCHEM	(5)N,(8)N-Deaza-(10)-N-methylfolate
Interactions	PUBCHEM	(S)-2-(4-(((2-(fluoromethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(ethyl((2-(hydroxymethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[2-[ethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-1,3-thiazole-5-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	9-chloro-3-methyl-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	2(R)-[4(S)-carboxy-4-[4-[N-(2-hydroxymethyl-4-oxo-4,6,7,8-tetrahydro-3H-cyclopenta[g]quinazolin-6-yl)-N-(2-propynyl)amino]-benzamido]butyramido]pentanedioic acid
Interactions	PUBCHEM	3-amino-9-bromo-10-nitrobenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	9-fluoro-3-methyl-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-hydroxy-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-7,9-dimethylbenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	9-Bromo-5,6-dihydro-3-methylbenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-9-vinylbenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-9-chloro-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3,10-diamino-9-bromobenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-9-ethynyl-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-8,9-dichloro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-methylsulfanyl-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3,8-diamino-9-bromo-5,6-dihydro-2H-benzo[f]quinazolin-1-one

Main Category	Category	Terms
Interactions	PUBCHEM	3-Amino-8,9-dichloro-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-hydroxy-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	7-chloro-3-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-8-chloro-6-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-methoxy-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-ethoxy-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	9-bromo-3-methylbenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-9-methoxy-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	N-(9-bromo-1-oxo-2H-benzo[f]quinazolin-3-yl)formamide
Interactions	PUBCHEM	3,8-diamino-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3,9-diaminobenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-9-bromo-5,6-dihydrobenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-7-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-8-bromo-5,6-dihydrobenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-6-(hydroxymethyl)-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-(benzylamino)-7-bromobenzo-[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-9-chloro-6-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	9-fluoro-3-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-iodo-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-ethylbenzo[f]-quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-9-fluoro-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-10-nitro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-ethylsulfanyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-6-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-iodo-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-fluoro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-Amino-5,6-dihydrobenzo[f]quinazolin-1(2H)-one
Interactions	PUBCHEM	3-amino-7-bromo-2H-benzo[f]quinazolin-1-one

Main Category	Category	Terms
Interactions	PUBCHEM	3,8,10-triamino-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-chloro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	9-iodo-3-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	9-chloro-3-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	9-iodo-3-methyl-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-methylsulfanyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-7-iodo-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-chloro-6-methyl-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-methyl-5,6-dihydro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	7-iodo-3-methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3,8-diamino-9-bromo-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-amino-9-bromo-8-nitro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	3-Methyl-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	2-[[4-[[(2-amino-4-oxo-3H-pyrido[2,3-d]pyrimidin-6-yl)amino]methyl]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[(2-amino-4-oxo-3H-pyrido[2,3-d]pyrimidin-6-yl)-methylamino]methyl]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(1,5-Dimethyl-2-methylimino-1,2-dihydro-benzo[cd]indol-6-yl)-methyl-[4-(morpholine-4-sulfonyl)-benzyl]-amine
Interactions	PUBCHEM	2-Amino-5-[(2,5-dimethoxyphenyl)sulfanyl]-6-methylthieno[2,3-d]pyrimidin-4(3H)-one
Interactions	PUBCHEM	2-Amino-6-methyl-5-(4-trifluoromethoxy-phenylsulfanyl)-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(4-chloro-3-nitro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(3-bromo-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(3-fluoro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-6-methyl-5-(4-trifluoromethyl-phenylsulfanyl)-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	4-(2-Amino-6-methyl-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-ylsulfanyl)-benzonitrile
Interactions	PUBCHEM	2-Amino-5-(3-chloro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(4-bromo-phenylsulfanyl)-6-methyl-3,7-

Main Category	Category	Terms
		dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(3,5-difluoro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(3-chloro-4-fluoro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(3,4-difluoro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-Amino-5-(3,5-dichloro-phenylsulfanyl)-6-methyl-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-{4-[(2-Amino-4-oxo-3,4-dihydro-quinazolin-6-ylmethyl)-cyanomethyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	2-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-propylamino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[4-[[4-[[2-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-ethynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[4-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-ethynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[2-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-ethynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[4-[[2-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-ethynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-ethynylamino]benzoyl]amino]-5-oxopentanoic acid
Interactions	PUBCHEM	2-[[4-[[2-amino-5-[(2,4-dioxo-1H-pyrimidin-5-yl)methyl]-4-oxo-3,6,7,8-tetrahydropteridin-6-yl]methyl-methylamino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[4-(3-Amino-1-oxo-1,2,5,6-tetrahydro-benzo[f]quinazoline-9-sulfonylamino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-{[4-(3-Amino-1-oxo-1,2,5,6-tetrahydro-benzo[f]quinazoline-8-sulfonylamino)-benzoyl]-prop-2-ynyl-amino}-pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-fluoro-4-(((4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	2-Amino-6-methyl-5-(naphthalen-2-ylsulfanyl)-3,7-dihydro-pyrrolo[2,3-d]pyrimidin-4-one

Main Category	Category	Terms
Interactions	PUBCHEM	6-{[(4-Benzenesulfinyl-3-fluoro-phenyl)-prop-2-ynyl-amino]-methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	6-({[4-(2-Chloro-benzenesulfinyl)-phenyl]-prop-2-ynyl-amino}-methyl)-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Methyl-6-({prop-2-ynyl-[4-(toluene-2-sulfonyl)-3-trifluoromethyl-phenyl]-amino}-methyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	6-{[(4-Benzenesulfonyl-3-chloro-phenyl)-prop-2-ynyl-amino]-methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Methyl-6-({prop-2-ynyl-[4-(toluene-2-sulfonyl)-phenyl]-amino}-methyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	6-{[(4-Benzenesulfonyl-3-fluoro-phenyl)-prop-2-ynyl-amino]-methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	6-{[(4-Benzenesulfinyl-3-trifluoromethyl-phenyl)-prop-2-ynyl-amino]-methyl}-2-methyl-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Methyl-6-({prop-2-ynyl-[4-(toluene-2-sulfinyl)-phenyl]-amino}-methyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	2-Methyl-6-({prop-2-ynyl-[4-(toluene-2-sulfinyl)-3-trifluoromethyl-phenyl]-amino}-methyl)-3H-quinazolin-4-one
Interactions	PUBCHEM	N-[[5-[[(3,4-Dihydro-2-methyl-4-oxo-6-quinazolinyl)methyl]ethylamino]-2-thienyl]carbonyl]-L-glutamic acid
Interactions	PUBCHEM	2-[4-(3-Methyl-1-oxo-1,2-dihydro-benzo[f]quinazoline-9-sulfonylamino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-[4-(2-{(2-Amino-4-methyl-6-oxo-1,6-dihydro-pyrimidin-5-yl)-[1-(4-hydroxy-5-phosphonooxymethyl-tetrahydro-furan-2-yl)-2,4-dioxo-1,2,3,4-tetrahydro-pyrimidin-5-ylmethyl]-amino}-ethylamino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-[4-({2-Amino-5-[1-(4-hydroxy-5-phosphonooxymethyl-tetrahydro-furan-2-yl)-2,4-dioxo-1,2,3,4-tetrahydro-pyrimidin-5-ylmethyl]-4-oxo-3,4,5,6,7,8-hexahydro-pyrido[3,2-d]pyrimidin-6-ylmethyl}-amino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-{4-[(2-{(2-Amino-4-methyl-6-oxo-1,6-dihydro-pyrimidin-5-yl)-[1-(4-hydroxy-5-phosphonooxymethyl-tetrahydro-furan-2-yl)-2,4-dioxo-1,2,3,4-tetrahydro-pyrimidin-5-ylmethyl]-amino}-ethyl)-methyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[4-[(2-amino-4-oxo-3H-pyrido[3,2-d]pyrimidin-6-yl)methyl-methylamino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[4-[(2-amino-4-oxo-3H-pyrido[3,2-d]pyrimidin-6-yl)methyl-prop-2-ynylamino]benzoyl]amino]pentanedioic acid

Main Category	Category	Terms
Interactions	PUBCHEM	(2S)-2-[[4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]-2-nitrobenzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	N-[[4-(benzenesulfonyl)phenyl]methyl]-N-methyl-2-methylimino-1H-benzo[cd]indol-6-amine
Interactions	PUBCHEM	N-[[4-(4-methoxyphenyl)sulfonylphenyl]methyl]-N,5-dimethyl-2-methylimino-1H-benzo[cd]indol-6-amine
Interactions	PUBCHEM	2-[4-(2-Amino-6-methyl-4-oxo-3,4-dihydro-quinazolin-5-ylsulfanyl)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-[4-(2,6-Dimethyl-4-oxo-3,4-dihydro-quinazolin-5-ylsulfanyl)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-{4-[(2-Amino-4-oxo-3,4-dihydro-pyrido[2,3-d]pyrimidin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	2-{4-[(2-Amino-4-oxo-3,4-dihydro-pyrido[3,2-d]pyrimidin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	(S)-2-(5-(methyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)picolinamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(6-(methyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)nicotinamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(5-(methyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)pyrimidine-2-carboxamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(5-(((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)picolinamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(5-(allyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)picolinamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(5-((2-fluoroethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)picolinamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(5-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)picolinamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((2-hydroxyethyl)((2-(hydroxymethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((2-isopropyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((2-mercaptoethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid

Main Category	Category	Terms
Interactions	PUBCHEM	(S)-2-(4-((3-hydroxypropyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((4-oxo-2-phenyl-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((2-bromoethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((2-(fluoromethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(allyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(allyl((2-(hydroxymethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((2-fluoroethyl)((2-(hydroxymethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((2-hydroxyethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((2-(acetamidomethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((2-(hydroxymethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((2-fluoroethyl)((2-(fluoromethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-fluoro-4-(((2-methoxy-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-fluoro-4-(((2-(hydroxymethyl)-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(((2-amino-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)-2-fluorobenzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-amino-4-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid

Main Category	Category	Terms
Interactions	PUBCHEM	(S)-2-(2-fluoro-4-((2-fluoroethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-amino-4-(methyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)-3-fluorobenzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-((cyanomethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)-2-fluorobenzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-fluoro-4-((2-hydroxyethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-amino-4-(((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(3-fluoro-4-((2-hydroxyethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(3-fluoro-4-((3-hydroxypropyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-fluoro-4-((3-hydroxypropyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-fluoro-4-((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methylamino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-chloro-4-(((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(allyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)-3-fluorobenzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(allyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)-2-fluorobenzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-chloro-4-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-methoxy-4-(((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(3-fluoro-4-((2-fluoroethyl)((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-

Main Category	Category	Terms
		yl)methyl)amino)-2-methoxybenzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-methyl-4-(methyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(3-fluoro-4-((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methylamino)benzamido)pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-(ethyl((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)amino)-2-methylbenzamido)pentanedioic acid
Interactions	PUBCHEM	2-{4-[Carbamoylmethyl-(2-methyl-4-oxo-3,4-dihydro-quinazolin-6-ylmethyl)-amino]-2-fluoro-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	(S)-2-(2-methyl-4-(((2-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)methyl)(prop-2-ynyl)amino)benzamido)pentanedioic acid
Interactions	PUBCHEM	2-[4-(2-Amino-6-methyl-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-ylsulfanyl)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-[4-(3-Methyl-1-oxo-1,2,5,6-tetrahydro-benzo[f]quinazoline-9-sulfonylamino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-[4-(3-Amino-8-bromo-1-oxo-1,2,5,6-tetrahydro-benzo[f]quinazoline-9-sulfonylamino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-[4-(3-Amino-1-oxo-1,2,5,6-tetrahydro-benzo[f]quinazoline-8-sulfonylamino)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[4-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[4-[[4-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[4-[[4-[[4-[[4-[(2-amino-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]-4-carboxybutanoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]thiophene-2-carbonyl]amino]pentanedioic acid

Main Category	Category	Terms
Interactions	PUBCHEM	(2S)-2-[[5-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-propylamino]thiophene-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-enylamino]thiophene-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[2-fluoroethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]thiophene-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[2-hydroxyethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]thiophene-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[3-hydroxypropyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]thiophene-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[cyanomethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]thiophene-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methylamino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[ethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-enylamino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[2-fluoroethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[2-hydroxyethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[5-[3-hydroxypropyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-1,3-thiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-Amino-4-oxo-5-thiopyridyl-6-methyl pyrrolo(2,3-d)pyrimidine
Interactions	PUBCHEM	(2S)-2-[[5-[ethyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]-1,3,4-thiadiazole-2-carbonyl]amino]pentanedioic acid
Interactions	PUBCHEM	6-[[N-ethyl-4-(2,2,2-trifluoroacetyl)anilino]methyl]-2-methyl-3H-quinazolin-4-one

Main Category	Category	Terms
Interactions	PUBCHEM	(Z)-but-2-enedioic acid;N-methyl-2-methylimino-N-[(4-morpholin-4-ylsulfonylphenyl)methyl]-1H-benzo[cd]indol-6-amine
Interactions	PUBCHEM	(2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-2-phenylacetic acid
Interactions	PUBCHEM	(2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-2-(3-hydroxyphenyl)acetic acid
Interactions	PUBCHEM	3-[(S)-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-carboxymethyl]benzoic acid
Interactions	PUBCHEM	methyl (2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-2-(3-hydroxyphenyl)acetate
Interactions	PUBCHEM	3-(benzylamino)-9-chloro-2H-benzo[f]quinazolin-1-one
Interactions	PUBCHEM	(2S)-2-[[3-fluoro-4-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl-prop-2-ynylamino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[3-fluoro-4-[methyl-[(2-methyl-4-oxo-3H-quinazolin-6-yl)methyl]amino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-{4-[(2-Amino-5-{1-[5-(dihydroxy-phosphanyloxymethyl)-4-hydroxy-tetrahydro-furan-2-yl]-2,4-dioxo-1,2,3,4-tetrahydro-pyrimidin-5-ylmethyl}-4-oxo-3,4,5,6,7,8-hexahydro-pyrido[3,2-d]pyrimidin-6-ylmethyl)-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	2-Amino-5-naphthalen-2-ylsulfanyl-3,9-dihydropyrimido[4,5-b]indol-4-one
Interactions	PUBCHEM	2-{(2z,5s)-4-Hydroxy-2-[(2e)-(2-Hydroxybenzylidene)hydrazinylidene]-2,5-Dihydro-1,3-Thiazol-5-Yl}-N-[3-(Trifluoromethyl)phenyl]acetamide
Interactions	PUBCHEM	5-(2,5-dimethoxyphenyl)sulfanyl-9H-pyrimido[4,5-b]indole-2,4-diamine
Interactions	PUBCHEM	5-naphthalen-2-ylsulfanyl-9H-pyrimido[4,5-b]indole-2,4-diamine
Interactions	PUBCHEM	5-naphthalen-1-ylsulfanyl-9H-pyrimido[4,5-b]indole-2,4-diamine
Interactions	PUBCHEM	5-(4-chlorophenyl)sulfanyl-9H-pyrimido[4,5-b]indole-2,4-diamine
Interactions	PUBCHEM	5-(4-methoxyphenyl)sulfanyl-9H-pyrimido[4,5-b]indole-2,4-diamine
Interactions	PUBCHEM	2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}amino)-4-chlorobenzoic acid
Interactions	PUBCHEM	[2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}

Main Category	Category	Terms
		amino)phenyl]acetic acid
Interactions	PUBCHEM	[2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}amino)-4-methoxyphenyl]acetic acid
Interactions	PUBCHEM	2-({4-[(2-amino-4-oxo-4,7-dihydro-1H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl} amino)benzoic acid
Interactions	PUBCHEM	2-[({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl} amino)methyl]benzoic acid
Interactions	PUBCHEM	3-[2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl} amino)phenyl]propanoic acid
Interactions	PUBCHEM	2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl} amino)benzene-1,4-dicarboxylic acid
Interactions	PUBCHEM	2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}amino)-4-methoxybenzoic acid
Interactions	PUBCHEM	2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}amino)-4-cyanobenzoic acid
Interactions	PUBCHEM	2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}amino)-4-carbamoylbenzoic acid
Interactions	PUBCHEM	[2-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}amino)-4-cyanophenyl]acetic acid
Interactions	PUBCHEM	3-({4-[(2-amino-4-oxo-4,7-dihydro-3H-pyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzene-1-carbonyl}amino)-4-(carboxymethyl)benzoic acid
Interactions	PUBCHEM	(2S)-2-[5-[[2-(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-6-yl)acetyl]amino]pentanoylamino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[4-[[2-(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-6-yl)acetyl]amino]butanoylamino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[4-[[2-(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-6-yl)acetyl]amino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[6-[[2-(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-6-yl)acetyl]amino]hexanoylamino]pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[3-[[2-(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-6-yl)acetyl]amino]propanoylamino]pentanedioic acid

Main Category	Category	Terms
Interactions	PUBCHEM	(2S)-2-[[2-[[2-(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-6-yl)acetyl]amino]acetyl]amino]pentanedioic acid
Interactions	PUBCHEM	5-(4-Benzenesulfonyl-benzyl)-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinolin-2-ylamine
Interactions	PUBCHEM	4-[4-(2-Amino-1,6,7,8-tetrahydro-imidazo[4,5-g]quinolin-5-ylmethyl)-benzenesulfonyl]-phenol
Interactions	PUBCHEM	5-[4-(4-Methoxy-benzenesulfonyl)-benzyl]-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinolin-2-ylamine
Interactions	PUBCHEM	[6-(2-Amino-1,6,7,8-tetrahydro-imidazo[4,5-g]quinolin-5-ylmethyl)-naphthalen-2-yl]-methanol
Interactions	PUBCHEM	5-[4-(Morpholine-4-sulfonyl)-benzyl]-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinolin-2-ylamine
Interactions	PUBCHEM	5-[4-(Piperazine-1-sulfonyl)-benzyl]-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinoline
Interactions	PUBCHEM	5-(4-Benzenesulfonyl-benzyl)-2-methylsulfanyl-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinoline
Interactions	PUBCHEM	[5-(4-Benzenesulfonyl-benzyl)-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinolin-2-yl]-methyl-amine
Interactions	PUBCHEM	5-[4-(4-Methoxy-benzenesulfonyl)-benzyl]-1-methyl-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinolin-2-ylamine
Interactions	PUBCHEM	5-[4-(4-Methoxy-benzenesulfonyl)-benzyl]-2-methyl-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinoline
Interactions	PUBCHEM	5-[4-(4-Methoxy-benzenesulfonyl)-benzyl]-1,2-dimethyl-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinoline
Interactions	PUBCHEM	5-[4-(benzenesulfonyl)phenyl]sulfanyl-5,6,7,8-tetrahydro-1H-benzo[f]benzimidazol-2-amine
Interactions	PUBCHEM	6-[[4-(benzenesulfonyl)phenyl]methylamino]-5-methyl-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[[4-(benzenesulfonyl)phenyl]methylamino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[[4-(benzenesulfonyl)phenyl]methyl-methylamino]-5-methyl-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	5-chloro-6-[methyl-[(4-morpholin-4-ylsulfonylphenyl)methyl]amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[methyl-[(4-morpholin-4-ylsulfonylphenyl)methyl]amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	5-methyl-6-[methyl-[(4-morpholin-4-ylsulfonylphenyl)methyl]amino]-1H-benzo[cd]indol-2-one

Main Category	Category	Terms
Interactions	PUBCHEM	(4-Benzenesulfonyl-benzyl)-(2-imino-1-methyl-1,2-dihydro-benzo[cd]indol-6-yl)-methyl-amine
Interactions	PUBCHEM	N-[[4-(benzenesulfonyl)phenyl]methyl]-2-hydrazinyl-N-methylbenzo[cd]indol-6-amine
Interactions	PUBCHEM	N-[6-[[4-(benzenesulfonyl)phenyl]methyl-methylamino]benzo[cd]indol-2-yl]hydroxylamine
Interactions	PUBCHEM	6-[(4-piperazin-1-ylsulfonylphenyl)methyl-propan-2-ylamino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[2-hydroxyethyl-[(4-piperazin-1-ylsulfonylphenyl)methyl]amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[3-hydroxypropyl-[(4-piperazin-1-ylsulfonylphenyl)methyl]amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[[4-(benzenesulfonyl)phenyl]methyl-(methylsulfanylmethyl)amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-{Methyl-[4-(piperazine-1-sulfonyl)-benzyl]-amino}-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	5-chloro-6-N-methyl-6-N-[(4-morpholin-4-ylsulfonylphenyl)methyl]benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	6-N-methyl-5-methylsulfanyl-6-N-[(4-morpholin-4-ylsulfonylphenyl)methyl]benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	6-[[4-(benzenesulfonyl)phenyl]methyl-ethylamino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[[4-(4-hydroxyphenyl)sulfonylphenyl]methyl-methylamino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[[4-(4-methoxyphenyl)sulfonylphenyl]methyl-methylamino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[methyl-[[4-(4-nitrophenyl)sulfonylphenyl]methyl]amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-N-[[4-(4-methoxyphenyl)sulfonylphenyl]methyl]-6-N-methylbenzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	4-[4-[[(2-Aminobenzo[cd]indol-6-yl)-methylamino]methyl]phenyl]sulfonylphenol
Interactions	PUBCHEM	6-[[4-(4-methoxyphenyl)sulfonylphenyl]methyl-methylamino]-5-methyl-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-N-[[4-(4-methoxyphenyl)sulfonylphenyl]methyl]-6-N,5-dimethylbenzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	N-[6-[[4-(4-methoxyphenyl)sulfonylphenyl]methyl-methylamino]-5-methylbenzo[cd]indol-2-yl]hydroxylamine
Interactions	PUBCHEM	5-methyl-6-[methyl-[[4-(4-

Main Category	Category	Terms
		nitrophenyl)sulfonylphenyl]methyl]amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-[[4-(4-aminophenyl)sulfonylphenyl]methyl-methylamino]-5-methyl-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	6-N,5-dimethyl-6-N-[[4-(4-nitrophenyl)sulfonylphenyl]methyl]benzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	6-N-[[4-(4-aminophenyl)sulfonylphenyl]methyl]-6-N,5-dimethylbenzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	6-N-[[4-(benzenesulfonyl)phenyl]methyl]-6-N-ethylbenzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	2-[[4-[(2-Amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzoyl]amino]-4-(hydroxymethyl)benzoic acid
Interactions	PUBCHEM	2-[[4-[(2-Amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzoyl]amino]cyclohexane-1-carboxylic acid
Interactions	PUBCHEM	4-[(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]-N-[2-(2H-tetrazol-5-yl)phenyl]benzamide
Interactions	PUBCHEM	2-[[4-[(2-Amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzoyl]amino]-3-(methoxymethyl)benzoic acid
Interactions	PUBCHEM	N-[2-[1-(3-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	2-[[4-[(2-Amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzoyl]amino]-5-methoxybenzoic acid
Interactions	PUBCHEM	N-[4-[1-(3-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	2-[[4-[(2-Amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzoyl]amino]-4-(methoxymethyl)benzoic acid
Interactions	PUBCHEM	N-[3-[1-(2-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	N-[4-[1-(4-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	N-[2-[1-(3-methylphenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	3-[[4-[(2-Amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzoyl]amino]benzoic acid
Interactions	PUBCHEM	(2S)-4-[1-(carboxymethyl)tetrazol-5-yl]-2-[[4-[(2,7-dimethyl-4-oxo-3H-quinazolin-6-yl)methyl-ethynylamino]-2-fluorobenzoyl]amino]butanoic acid
Interactions	PUBCHEM	N-[3-[1-(4-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-

Main Category	Category	Terms
		pyrimidine-5-sulfonamide
Interactions	PUBCHEM	N-[2-[1-(2-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	4-[[4-[(2-Amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]benzoyl]amino]benzoic acid
Interactions	PUBCHEM	N-[2-[1-(4-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	2-[(E)-2-[4-[(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]phenyl]ethenyl]benzoic acid
Interactions	PUBCHEM	4-[(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)methyl]-N-(2-cyanophenyl)benzamide
Interactions	PUBCHEM	N-[3-[1-(3-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	N-[4-[1-(2-chlorophenyl)triazol-4-yl]phenyl]-2,4-dioxo-1H-pyrimidine-5-sulfonamide
Interactions	PUBCHEM	N-4-hydroxy-2'-deoxycytidylic acid
Interactions	PUBCHEM	3,3-Bis(4-hydroxyphenyl)benzo[g]isobenzofuran-1-one
Interactions	PUBCHEM	5-(4-Benzoquinonyl)-2'-deoxyuridine 5'-phosphate
Interactions	PUBCHEM	Ethyl 5-phenyl-1-[1-[3-[(2,2,2-trifluoroacetyl)amino]propyl]triazole-4-carbonyl]pyrrolidine-2-carboxylate
Interactions	PUBCHEM	3-[4-[(5-Methyl-2-propan-2-ylphenoxy)methyl]triazol-1-yl]benzoic acid
Interactions	PUBCHEM	1-(2-Methylphenyl)-4-[(5-methyl-2-propan-2-ylphenoxy)methyl]triazole
Interactions	PUBCHEM	2-Phenyl-5-[(1-phenyltriazol-4-yl)methylsulfanyl]-1,3,4-oxadiazole
Interactions	PUBCHEM	5-methyl-1-phenyl-N-(2-(thiophen-2-yl)ethyl)-1H-1,2,3-triazole-4-carboxamide
Interactions	PUBCHEM	methyl (2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-3-(4-hydroxyphenyl)propanoate
Interactions	PUBCHEM	methyl 3-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]benzoate
Interactions	PUBCHEM	methyl (2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-3-phenylpropanoate
Interactions	PUBCHEM	methyl (2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-3-(3,4-dihydroxyphenyl)propanoate
Interactions	PUBCHEM	methyl (2S)-2-[[5-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]thiophene-2-carbonyl]amino]-2-phenylacetate

Main Category	Category	Terms
Interactions	PUBCHEM	3-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]benzoic acid
Interactions	PUBCHEM	(2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-3-phenylpropanoic acid
Interactions	PUBCHEM	4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]-N-[(1S)-2-hydroxy-1-phenylethyl]benzamide
Interactions	PUBCHEM	4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]-N-[(1R)-1-phenylethyl]benzamide
Interactions	PUBCHEM	methyl (2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-3-(1H-indol-3-yl)propanoate
Interactions	PUBCHEM	methyl (2R)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-2-phenylacetate
Interactions	PUBCHEM	methyl (2S)-2-[[4-[(2-amino-6-methyl-4-oxo-3H-quinazolin-5-yl)sulfanyl]benzoyl]amino]-3-(4-aminophenyl)propanoate
Interactions	PUBCHEM	2',5-Difluoro-1-arabinosyluridine monophosphate
Interactions	PUBCHEM	1,8-Naphthalein derivative, 12
Interactions	PUBCHEM	1,8-Naphthalein derivative, 14
Interactions	PUBCHEM	1,8-Naphthalein derivative, 15
Interactions	PUBCHEM	2-[[2-Propynyl(3,4-difluorophenyl)amino]methyl]-3-phenyl-7-t rfluoromethylquinoxaline
Interactions	PUBCHEM	2-[[2-Propynyl(4-fluorophenyl)amino]methyl]-3-phenyl-7-trifl uoromethylquinoxaline
Interactions	PUBCHEM	2-[[2-Propynyl(4-methylphenyl)amino]methyl]-3-phenyl-7-trifl uoromethylquinoxaline
Interactions	PUBCHEM	2-[(3,4-Dichloro-phenyl)amino]-3-phenyl-5,7-diaminoquinoxaline
Interactions	PUBCHEM	2-[[2-Methyl(3,4-difluorophenyl)amino]methyl]-3-phenyl-7-tri fluoromethylquinoxaline
Interactions	PUBCHEM	2-[(4-Methoxy-phenyl)amino]-3-phenyl-5,7-dinitroquinoxaline
Interactions	PUBCHEM	2-Amino-4-oxo-6-methyl-5-phenylsulfanylthieno[2,3-d]pyrimidine
Interactions	PUBCHEM	2-Amino-5-[(4-chlorophenyl)sulfanyl]-6-methylthieno[2,3-d]pyrimidin-4(3H)-one
Interactions	PUBCHEM	2-Amino-5-[(3,5-dichlorophenyl)sulfanyl]-6-methylthieno[2,3-d]pyrimidin-4(3H)-one
Interactions	PUBCHEM	2-Amino-6-methyl-5-(pyridin-4-ylsulfanyl)thieno[2,3-d]pyrimidin-4(3H)-one
Interactions	PUBCHEM	2-Amino-5-[(3,4-dichlorophenyl)sulfanyl]-6-methylthieno[2,3-d]pyrimidin-4(3H)-one

Main Category	Category	Terms
Interactions	PUBCHEM	2-Amino-6-methyl-5-(2-naphthylsulfanyl)thieno[2,3-d]pyrimidin-4(3H)-one
Interactions	PUBCHEM	2-Amino-5-[(4-fluorophenyl)sulfanyl]-6-methylthieno[2,3-d]pyrimidin-4(3H)-one
Interactions	PUBCHEM	5-(phenylthio)-9H-pyrimido[4,5-b]indole-2,4-diamine
Interactions	PUBCHEM	5-((4-methylphenyl)thio)-9H-pyrimido(4,5-b)indole-2,4-diamine
Interactions	PUBCHEM	Metesind
Interactions	PUBCHEM	1,3-dioxo-1H,3H-benzo[de]isochromene-6-carboxylic acid
Interactions	PUBCHEM	2-[4-(Methoxy)anilino]-3-phenyl-5,7-diammino quinoxaline
Interactions	PUBCHEM	3,4,5-trimethoxy-N-[[3-phenyl-7-(trifluoromethyl)quinoxalin-2-yl]methyl]-N-prop-2-ynylaniline
Interactions	PUBCHEM	6-Methyl-2,3-dihydro-phenalen-1-one
Interactions	PUBCHEM	1,3-Propylenediphosphonic Acid
Interactions	PUBCHEM	N-({4-[(2-Amino-6-Ethyl-4-Oxo-3,4-Dihydrothieno[2,3-D]pyrimidin-5-Yl)sulfanyl]phenyl}carbonyl)-L-Glutamic Acid
Interactions	PUBCHEM	2-amino-5-(4-chlorophenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-(2,5-dimethoxyphenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-(3,5-dichlorophenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-6-ethyl-5-(4-fluorophenyl)sulfanyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-(3-chlorophenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-(2-chlorophenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-6-ethyl-5-phenylsulfanyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-6-ethyl-5-(4-nitrophenyl)sulfanyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-(3,4-dichlorophenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-6-ethyl-5-(2-naphthylsulfanyl)-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-6-ethyl-5-(4-pyridylsulfanyl)-3H-thieno[2,3-d]pyrimidin-4-one

Main Category	Category	Terms
Interactions	PUBCHEM	2-amino-5-(4-bromophenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-amino-5-(3,5-dimethoxyphenyl)sulfanyl-6-ethyl-3H-thieno[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	2-[[4-[[(2-methyl-4-oxo-3H-quinolin-6-yl)methyl-prop-2-ynylamino]methyl]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[(2R)-2-methyl-4-oxo-2,3-dihydro-1H-quinazolin-6-yl]methyl-prop-2-ynylamino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-[[4-[[(1-oxo-2H-isoquinolin-7-yl)methyl-prop-2-ynylamino]methyl]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	(S)-2-{[5-(5-Fluoro-2,4-dioxo-3,4-dihydro-2H-pyrimidin-1-yl)-3-hydroxy-tetrahydro-furan-2-ylmethoxy]-hydroxy-phosphorylamino}-3-phenyl-propionic acid methyl ester
Interactions	PUBCHEM	2-[[4-[(2-Amino-6-methyl-4-oxo-3,4a,7,7a-tetrahydropyrrolo[2,3-d]pyrimidin-5-yl)sulfanyl]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	2-Amino-6-methyl-5-pyridin-4-ylsulfanyl-3,4a,7,7a-tetrahydropyrrolo[2,3-d]pyrimidin-4-one
Interactions	PUBCHEM	1-[8-Amino-5-[[4-(4-aminophenyl)sulfonylphenyl]methyl-methylamino]naphthalen-1-yl]ethanone
Interactions	PUBCHEM	2-{4-[(2,4-Diamino-pyrido[3,2-d]pyrimidin-6-ylmethyl)-prop-2-ynyl-amino]-benzoylamino}-pentanedioic acid
Interactions	PUBCHEM	4-[4-[[Methyl-(2-methylsulfanyl-1,2-dihydrobenzo[cd]indol-6-yl)amino]methyl]phenyl]sulfonylphenol
Interactions	PUBCHEM	6-N-[[4-(benzenesulfonyl)phenyl]methyl]-6-N,5-dimethylbenzo[cd]indole-2,6-diamine
Interactions	PUBCHEM	2-[4-(2-Amino-1,6,7,8-tetrahydro-imidazo[4,5-g]quinolin-5-ylmethyl)-benzoylamino]-pentanedioic acid
Interactions	PUBCHEM	(S)-2-(4-Benzyloxycarbonylamino-naphthalene-1-sulfonylamino)-3-[4-(5-dimethylamino-naphthalene-1-sulfonyloxy)-phenyl]-propionic acid
Interactions	PUBCHEM	(S)-3-[4-(5-Dimethylamino-naphthalene-1-sulfonyloxy)-phenyl]-2-(6-nitro-naphthalene-1-sulfonylamino)-propionic acid
Interactions	PUBCHEM	(S)-2-(6-Amino-naphthalene-1-sulfonylamino)-3-[4-(5-dimethylamino-naphthalene-1-sulfonyloxy)-phenyl]-propionic acid
Interactions	PUBCHEM	(S)-3-[4-(5-Dimethylamino-naphthalene-1-sulfonyloxy)-phenyl]-2-(1-oxa-2,3-diaza-cyclopenta[a]naphthalene-5-sulfonylamino)-propionic acid
Interactions	PUBCHEM	(S)-2-(5-Benzyloxycarbonylamino-naphthalene-1-sulfonylamino)-3-[4-(5-dimethylamino-naphthalene-1-sulfonyloxy)-phenyl]-propionic acid

Main Category	Category	Terms
Interactions	PUBCHEM	3-[[4-[(2-amino-6-methyl-4-oxo-3H-thieno[2,3-d]pyrimidin-5-yl)sulfanyl]benzoyl]-carboxyamino]propanoic acid
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)benzenesulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)-3-nitrobenzenesulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)thiophene-2-sulfonamide
Interactions	PUBCHEM	(2S)-2-{[4-(2-{2-amino-4-oxo-1H,4H,7H-pyrrolo[2,3-d]pyrimidin-5-yl}ethyl)phenyl]formamido}pentanedioic acid
Interactions	PUBCHEM	(2S)-2-[[4-(2,4-diamino-5-methylfuro[2,3-d]pyrimidin-6-yl)sulfanylbenzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	Phenolphthalein
Interactions	PUBCHEM	6-[ethyl-[(4-piperazin-1-ylsulfonylphenyl)methyl]amino]-1H-benzo[cd]indol-2-one;hydrochloride
Interactions	PUBCHEM	n-[4-[2-(2-Amino-4-methyl-7h-pyrrolo[2,3-d]pyrimidin-5-yl)-ethyl]-benzoyl]glutamic acid
Interactions	PUBCHEM	2-(4-Tert-butylphenyl)-5-[2-(2-methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-1,3,4-oxadiazole
Interactions	PUBCHEM	2-[2-(2-Methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-5-(4-propan-2-ylphenyl)-1,3,4-oxadiazole
Interactions	PUBCHEM	2-(4-Ethoxyphenyl)-5-[2-(2-methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-1,3,4-oxadiazole
Interactions	PUBCHEM	2-[2-(2-Methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-5-(4-methylphenyl)-1,3,4-oxadiazole
Interactions	PUBCHEM	N-[(4-{[(2-Amino-4-Oxo-3,4-Dihydro[1]benzothieno[2,3-D]pyrimidin-5-Yl)methyl]amino}phenyl)carbonyl]-L-Glutamic Acid
Interactions	PUBCHEM	(2s)-2-(5-{[(2-Amino-4-Oxo-3,4-Dihydro[1]benzothieno[2,3-D]pyrimidin-5-Yl)methyl]amino}-1-Oxo-1,3-Dihydro-2h-Isoindol-2-Yl)pentanedioic Acid
Interactions	PUBCHEM	N'-[(5-ethyl-2-thienyl)methylene]-1H-indole-3-carbohydrazide
Interactions	PUBCHEM	5-((4-Methyl-1,2,3,4-tetrahydroquinoxalyl)methyl)-2'-deoxyuridine 5'-phosphate
Interactions	PUBCHEM	(Z)-but-2-enedioic acid;6-N,5-dimethyl-6-N-[(4-morpholin-4-ylsulfonylphenyl)methyl]benzo[cd]indole-2,6-diamine

Main Category	Category	Terms
Interactions	PUBCHEM	(Z)-but-2-enedioic acid;6-[methyl-[(4-piperazin-1-ylsulfonylphenyl)methyl]amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	(2S)-2-[[4-[(2-methyl-4-oxo-3H-[1]benzothiolo[2,3-d]pyrimidin-5-yl)methylamino]benzoyl]amino]pentanedioic acid
Interactions	PUBCHEM	[(2R,3S,5R)-5-(5-fluoro-2,4-dioxopyrimidin-1-yl)-3-hydroxyoxolan-2-yl]methoxy-N-[(1S)-1-(3H-indol-3-yl)-2-methoxy-2-oxoethyl]phosphonamidic acid
Interactions	PUBCHEM	(2S)-2-[6-[(2-methyl-4-oxo-3H-[1]benzothiolo[2,3-d]pyrimidin-5-yl)methylamino]-3-oxo-1H-isoindol-2-yl]pentanedioic acid
Interactions	PUBCHEM	N-[(1E)-3-{[2-(1H-indol-3-yl)ethyl]amino}-3-oxo-1-(thiophen-2-yl)prop-1-en-2-yl]thiophene-2-carboxamide
Interactions	PUBCHEM	2H-Naphtho[1,8-bc]thiophen-2-one
Interactions	PUBCHEM	5-(3,3,3-Trifluoro-1-propenyl)-2'-deoxyuridylate
Interactions	PUBCHEM	dUMP
Interactions	PUBCHEM	Naphthalic anhydride
Interactions	PUBCHEM	5-(benzenesulfonyl)-N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)thiophene-2-sulfonamide
Interactions	PUBCHEM	2,5-dichloro-N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)thiophene-3-sulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)naphthalene-1-sulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)quinoline-8-sulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)-2,1,3-benzothiadiazole-4-sulfonamide
Interactions	PUBCHEM	(E)-N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)-2-phenylethenesulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)-5-[1-methyl-5-(trifluoromethyl)pyrrol-3-yl]thiophene-2-sulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)-3-nitrobenzamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-

Main Category	Category	Terms
		pentaen-7-yl)-4-nitrobenzenesulfonamide
Interactions	PUBCHEM	N-(2,4-dioxo-3-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8,10-pentaen-7-yl)-2-nitro-4-(trifluoromethyl)benzenesulfonamide
Interactions	PUBCHEM	2-[2-(2-Methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-5-(2-nitrophenyl)-1,3,4-oxadiazole
Interactions	PUBCHEM	2-(4-Fluorophenyl)-5-[2-(2-methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-1,3,4-oxadiazole
Interactions	PUBCHEM	2-[2-(2-Methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-5-(3,4,5-trimethoxyphenyl)-1,3,4-oxadiazole
Interactions	PUBCHEM	3-[5-[2-(2-Methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-1,3,4-oxadiazol-2-yl]pyridin-4-amine
Interactions	PUBCHEM	4-[5-[2-(2-Methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-1,3,4-oxadiazol-2-yl]pyridin-2-amine
Interactions	PUBCHEM	2-(2-Chloro-4-pyridinyl)-5-[2-(2-methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-1,3,4-oxadiazole
Interactions	PUBCHEM	2-(2,3-Dihydro-1,4-benzodioxin-6-yl)-5-[2-(2-methyl-5-nitroimidazol-1-yl)ethylsulfanyl]-1,3,4-oxadiazole
Interactions	PUBCHEM	(3-Oxo-2-oxatricyclo[6.3.1.04,12]dodeca-1(12),4,6,8,10-pentaen-9-yl) 3-cyanobenzoate
Interactions	PUBCHEM	(3-Oxo-2-oxatricyclo[6.3.1.04,12]dodeca-1(12),4,6,8,10-pentaen-9-yl) prop-2-enoate
Interactions	PUBCHEM	1-(4-Acetylpiperazin-1-yl)-2-[2-imino-4-(4-methylphenyl)-1,3-thiazol-3-yl]ethanone
Interactions	PUBCHEM	2-(2-Imino-4-phenyl-1,3-thiazol-3-yl)-1-morpholin-4-ylethanone
Interactions	PUBCHEM	2-[4-(4-Chlorophenyl)-2-imino-1,3-thiazol-3-yl]-1-morpholin-4-ylethanone
Interactions	PUBCHEM	2-[4-(4-Chlorophenyl)-2-imino-1,3-thiazol-3-yl]-1-(4-methylpiperazin-1-yl)ethanone
Interactions	PUBCHEM	2-(2-Imino-4-phenyl-1,3-thiazol-3-yl)-1-(4-methylpiperazin-1-yl)ethanone
Interactions	PUBCHEM	2-[2-Imino-4-(4-methylphenyl)-1,3-thiazol-3-yl]-1-(4-phenylpiperazin-1-yl)ethanone
Interactions	PUBCHEM	2-(2-Imino-4-phenyl-1,3-thiazol-3-yl)-1-(4-phenylpiperazin-1-yl)ethanone
Interactions	PUBCHEM	2-[2-Imino-4-(4-methylphenyl)-1,3-thiazol-3-yl]-1-morpholin-4-ylethanone
Interactions	PUBCHEM	2-[4-(4-Chlorophenyl)-2-imino-1,3-thiazol-3-yl]-1-(4-phenylpiperazin-1-yl)ethanone
Interactions	PUBCHEM	US9422297, Pddf

Main Category	Category	Terms
Interactions	PUBCHEM	1H,3H-Naphtho[1,8-cd]pyran-1,3-dione, 6-chloro-
Interactions	PUBCHEM	2H-Naphtho[1,8-bc]furan-2-one
Interactions	PUBCHEM	5-Fluorodeoxyuridine monophosphate
Interactions	PUBCHEM	4-Chloro-1-phenyl-[1,2,4]triazolo[4,3-a]quinoxaline
Interactions	PUBCHEM	[(2R,3S,5R)-3-hydroxy-5-[5-(4-nitrophenyl)-2,4-dioxopyrimidin-1-yl]oxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-(2,4-dioxo-5-phenylpyrimidin-1-yl)-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-3-hydroxy-5-[5-(4-methylphenyl)-2,4-dioxopyrimidin-1-yl]oxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-3-hydroxy-5-[5-(4-methoxyphenyl)-2,4-dioxopyrimidin-1-yl]oxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-[5-(4-aminophenyl)-2,4-dioxopyrimidin-1-yl]-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-[2,4-dioxo-5-(4-phenylmethoxyphenyl)pyrimidin-1-yl]-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-3-hydroxy-5-[5-(4-hydroxyphenyl)-2,4-dioxopyrimidin-1-yl]oxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-[5-[4-(dimethylamino)phenyl]-2,4-dioxopyrimidin-1-yl]-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-[5-(4-cyanophenyl)-2,4-dioxopyrimidin-1-yl]-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-[2,4-dioxo-5-[4-(trifluoromethyl)phenyl]pyrimidin-1-yl]-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-[5-(4-carbamoylphenyl)-2,4-dioxopyrimidin-1-yl]-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	[(2R,3S,5R)-5-[5-(4-bromophenyl)-2,4-dioxopyrimidin-1-yl]-3-hydroxyoxolan-2-yl]methyl dihydrogen phosphate
Interactions	PUBCHEM	4-[1-(4-Hydroxy-5-phosphonooxymethyl-tetrahydro-furan-2-yl)-2,4-dioxo-1,2,3,4-tetrahydro-pyrimidin-5-yl]-benzoic acid methyl ester
Interactions	PUBCHEM	6-[(4-Benzenesulfonyl-benzyl)-methyl-amino]-1H-benzo[cd]indol-2-one
Interactions	PUBCHEM	N*6*-Methyl-N*6*-pyridin-4-ylmethyl-benzo[cd]indole-2,6-diamine

Main Category	Category	Terms
Interactions	PUBCHEM	5-[4-(Piperazine-1-sulfonyl)-benzyl]-5,6,7,8-tetrahydro-1H-imidazo[4,5-g]quinolin-2-ylamine
Interactions	PUBCHEM	N*6*-Methyl-N*6*-pyridazin-4-ylmethyl-benzo[cd]indole-2,6-diamine
Literature	GENERIF_SUMMARY	liganded crystal structures with pyrrolo(2,3-d)pyrimidine-based antifolate compounds
Literature	GENERIF_SUMMARY	high level of expression associated with poor prognosis in p-stage I adenocarcinoma of the lung
Literature	GENERIF_SUMMARY	Length polymorphism of thymidylate synthase regulatory region in Chinese populations and evolution of the novel alleles
Literature	GENERIF_SUMMARY	TK1 gene expression together with TS, TP and DPD gene expression may play important roles in influencing the malignant behavior of epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	Mutants of TS that confer resistance to MTX as well as 5-FU by random as well as site-directed mutagenesis. Review.
Literature	GENERIF_SUMMARY	Thymidylate synthase as a translational regulator of cellular gene expression. Review.
Literature	GENERIF_SUMMARY	rTS gene expression likely plays a role in down-regulating TS through a natural RNA-based antisense mechanism. Review.
Literature	GENERIF_SUMMARY	Induction of thymidylate synthase as a 5-fluorouracil resistance mechanism. Review.
Literature	GENERIF_SUMMARY	Structure-based studies on species-specific inhibition of thymidylate synthase. Review.
Literature	GENERIF_SUMMARY	mutations in TS result in conformational changes that confer 5-fluordeoxyuridine resistance
Literature	GENERIF_SUMMARY	TYMS and MTHFR compete for limiting supplies of folate required for the remethylation of homocysteine.
Literature	GENERIF_SUMMARY	the identified highly conserved protein domains, which occur at the homodimeric interface of thymidylate synthase(TS), represent potential participating sites for binding of TS protein to its mRNA.
Literature	GENERIF_SUMMARY	Enhanced expression mediates resistance of uterine cervical cancer cells to radiation
Literature	GENERIF_SUMMARY	cytomegalovirus infection induces cellular thymidylate synthase gene expression in quiescent human embryonic lung fibroblasts
Literature	GENERIF_SUMMARY	Gene expression is a predictor for the efficacy of fluoropyrimidine-based chemotherapy for metastatic colorectal cancer.
Literature	GENERIF_SUMMARY	Expression predicts the reponse of metastatic colorectal cancer to raltitrexel.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Associations between polymorphisms in the thymidylate synthase and serine hydroxymethyltransferase genes and susceptibility to malignant lymphoma.
Literature	GENERIF_SUMMARY	polymorphism and overexpression of Thymidylate synthase mrna is associated with non-small-cell lung cancer
Literature	GENERIF_SUMMARY	Results suggest that the genotyping for thymidylate synthase and methylenetetrahydrofolate reductase polymorphisms may be a useful indicator in determining the appropriate dose of methotrexate in patients with rheumatoid arthritis.
Literature	GENERIF_SUMMARY	Overexpression of thymidylate synthase mediates desensitization for 5-fluorouracil of cervical cancer cells
Literature	GENERIF_SUMMARY	Results describe the kinetic parameters of human recombinant thymidylate synthase (hrTS) with its natural substrate, dUMP, and E-5-(2-bromovinyl)-2(')-deoxyuridine monophosphate (BVdUMP).
Literature	GENERIF_SUMMARY	Findings identify Cys-180 as a critical residue for the in vitro and in vivo translational regulatory effects of human thymidylate synthase.
Literature	GENERIF_SUMMARY	TS and DPD quantitation may be helpful to evaluate prognosis of patients receiving adjuvant 5-FU and that patients with high TS and low DPD may benefit from adjuvant 5-FU chemotherapy in colorectal cancer.
Literature	GENERIF_SUMMARY	High TS expression is a marker of poor prognosis in resected pancreatic cancer. Patients with high intratumoral TS expression benefit from adjuvant therapy.
Literature	GENERIF_SUMMARY	These results indicate that postoperative chemotherapy with 5-FU could be effective for patients with the G-phenotype tumor, since the incidence of intratumoral expression of TS, the target enzyme of 5-FU, is significantly low in G-phenotype tumors
Literature	GENERIF_SUMMARY	There is prognostic significance to TS mRNA expression in breast neoplasms.
Literature	GENERIF_SUMMARY	thymidylate synthase loss of heterozygosity has a role in progression of colorectal cancer
Literature	GENERIF_SUMMARY	Molecular features of thymidylate synthase that control its degradation are examined; the carboxyl-terminal conformational shift is not required for ligand-mediated stabilization; the amino-terminus governs TS stability and its response to ligands.
Literature	GENERIF_SUMMARY	TYMS gene amplification in 23% of 31 5-FU-treated cancers, whereas no amplification was observed in metastases of patients that had not been treated with 5-FU.
Literature	GENERIF_SUMMARY	Over expression of Thymidylate synthase exhibits oncogene-like activity and suggest a link between DNA

Main Category	Category	Terms
		synthesis and the induction of a neoplastic phenotype
Literature	GENERIF_SUMMARY	6 bp/1494 polymorphism varies greatly within different ethnic populations and is in linkage disequilibrium with the thymidylate synthase 5' tandem repeat enhancer polymorphism
Literature	GENERIF_SUMMARY	thymidylate synthase and p53 have roles in regulating Fas-mediated apoptosis in response to antimetabolites
Literature	GENERIF_SUMMARY	Polymorphism of TYMS partially modifies the risk of esophageal and stomach cancer in patients with smoking habit.
Literature	GENERIF_SUMMARY	Thymidylate synthase inhibition triggers glucose-dependent apoptosis in p53-negative leukemic cells
Literature	GENERIF_SUMMARY	The combination of low expression of TS and induction of p16(INK4a) after chemotherapy can be important indicators of the sensitivity to 5-FU-based chemotherapy in colorectal cancers.
Literature	GENERIF_SUMMARY	Thymidylate synthase polymorphisms are associated with esophageal squamous cell carcinoma and gastric cardiac adenocarcinoma
Literature	GENERIF_SUMMARY	TS in malignant tissue was statistically higher than in normal stromal tissue in both differentiated and undifferentiated types (p
Literature	GENERIF_SUMMARY	Polymorphisms in the TS gene may contribute to gastric cancer susceptibility.
Literature	GENERIF_SUMMARY	Polymorphisms of reduced folate carrier,aminoimidazole carboxamide ribonucleotide transformylase,and thymidylate synthase genes contribute to the therapeutic response in rheumatoid arthritis patients to methotrexate.
Literature	GENERIF_SUMMARY	The common MTHFR C677T and TS enhancer region polymorphisms were not risk factors for breast cancer in this patient cohort nor were they associated with phenotypic features or with prognosis.
Literature	GENERIF_SUMMARY	Thymidylate synthase polymorphism may have a role in colorectal cancer
Literature	GENERIF_SUMMARY	there is a high level of homology in polymorphic status of thymidylate synthase between normal and malignant colonic tissues
Literature	GENERIF_SUMMARY	thymidylate synthase is inhibited by trifluorothymidine in human tumor cells
Literature	GENERIF_SUMMARY	Thymidylate synthase 5'- and 3'-untranslated region polymorphisms are associated with risk and progression of squamous cell carcinoma of the head and neck
Literature	GENERIF_SUMMARY	Our findings suggest that the TYMS 3'UTR del/del genotype is a significant determinant of elevated RBC folate concentration in a non-smoking population

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Investigators found a relationship between the TSmRNA expression, evaluated by real-time PCR, and with the TS level determined by immunohistochemical assay in colorectal cancer.
Literature	GENERIF_SUMMARY	High TS1 gene expression is associated with chemotherapy resistance in esophageal cancer
Literature	GENERIF_SUMMARY	Important role for folate deficiency and impaired TS activity in the etiology of esophagel and gastric cancer.
Literature	GENERIF_SUMMARY	Polymorphisms in the 5'- and 3'-untranslated regions are a risk factor for gastric cancer in China.
Literature	GENERIF_SUMMARY	The nasopharyngeal carcinoma patients with positive LRP and TS expression may be less sensitive to chemotherapy with DDP + 5-FU.
Literature	GENERIF_SUMMARY	TS polymorphism may be a genetic determinant of plasma homocysteine level in Korean patients with recurrent spontaneous abortion.
Literature	GENERIF_SUMMARY	thymidylate synthase and dihydropyrimidine dehydrogenase are involved in resistance to 5-fluorouracil in human lung cancer cells
Literature	GENERIF_SUMMARY	thymidylate synthase and orotate phosphoribosyl transferase, but not dihydropyrimidine dehydrogenase, are more highly expressed in prostate cancer than in benign prostatic hyperplasia
Literature	GENERIF_SUMMARY	Thymidylate synthase expression levels might be used both as a prognostic marker and to help identify patients who would benefit from 5-FU based regime.
Literature	GENERIF_SUMMARY	analysis of TS polymorphisms in childhood acute lymphoblastic leukemia patients
Literature	GENERIF_SUMMARY	Association of mRNA expresion patterns with tumor stage and suggested new prognostic and predictive markers for patients with colorectal cancer.
Literature	GENERIF_SUMMARY	the rTS signaling pathway may be active within human mitochondria
Literature	GENERIF_SUMMARY	This study demonstrates the ubiquitin-independent nature of TS proteolysis by showing that a 'lysine-less' polypeptide, in which all lysine residues were replaced by arginine, is still subject to proteasome-mediated degradation.
Literature	GENERIF_SUMMARY	The Thymidylate Synthase expression increased significantly between baseline and biopsy 2 and dropped to near baseline levels at biopsy 3.
Literature	GENERIF_SUMMARY	Expression level and polymorphisms predict disease-free survival in 5-fluorouracil treated colorectal cancer.
Literature	GENERIF_SUMMARY	The Thymidylate Synthase(TS) mRNA in plasma originated from tumors, it may indicate poor prognosis and might help to classify tumors in Dukes' stages B and C. The TS promoter enhancer region genotype may

Main Category	Category	Terms
		influence TS mRNA expression in plasma.
Literature	GENERIF_SUMMARY	Expression is not prognostic of survival in colon cancer.
Literature	GENERIF_SUMMARY	RTQ-LDA and RTQ identified thymidylate synthetase as significantly over-expressed in dead-of-disease compared to long-term survival glioblastoma multiforme patients.
Literature	GENERIF_SUMMARY	TS tandem repeat polymorphism was significantly related to survival.
Literature	GENERIF_SUMMARY	5FU sensitivity of a tumor is modified by aneuploidy-producing copy-number changes of TS alleles by one or more of the following
Literature	GENERIF_SUMMARY	analysis of Ki67 and thymidylate synthase expression in primary tumour compared with metastatic nodes in breast cancer patients
Literature	GENERIF_SUMMARY	thymidylate synthase and dihydropyrimidine dehydrogenase gene polymorphisms have a role in response and toxicity to capecitabine-raltitrexed treatment in colorectal cancer
Literature	GENERIF_SUMMARY	Findings provide fundamental and useful information for genotyping TYMS in the Japanese and probably other Asian populations.
Literature	GENERIF_SUMMARY	genetic polymorphisms of TS as a prognostic factor in advanced colorectal cancer patients treated with 5-fluorouracil plus oxaliplatin or irinotecan.
Literature	GENERIF_SUMMARY	a significant association between the presence of the 3G allele and TYMS mRNA expression and catalytic activity, particularly in p53-mutated cell lines
Literature	GENERIF_SUMMARY	increased risk of breast cancer associated with the double repeat (2R) allele in the thymidylate synthase (TYMS) 5'-untranslated region (UTR) (P, trend = 0.07)
Literature	GENERIF_SUMMARY	Gene expression may be a response to fluorouracil sensitivity in stomach cancers of different histologic origin.
Literature	GENERIF_SUMMARY	thymidylate synthetase 1494del6 polymorphism is associated with a significantly lower risk of breast cancer and may be a potential genetic marker
Literature	GENERIF_SUMMARY	No differences of these polymorphisms were found between ALL cells and normal blood cells, indicating an ethnic rather than leukemic origin.
Literature	GENERIF_SUMMARY	findings suggest that there may be an important relationship between the TYMS haplotypes examined and 5-fluorouracil toxicity
Literature	GENERIF_SUMMARY	similar frequencies of the MTHFR, the MTRR and the TYMS genotypes were seen in patients and controls.
Literature	GENERIF_SUMMARY	This study supports the hypothesis that reduced MTHFR activity and enhanced TYMS activity, both of

Main Category	Category	Terms
		which are essential elements in minimizing uracil misincorporation into DNA, may protect against the development of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Patients with thymidylate synthase genotype 2/2 had a significantly higher rate of complete pathologic response with 53 percent (8/15) compared with 22 percent in the 2/3 or 3/3 group.
Literature	GENERIF_SUMMARY	study found that functional polymorphisms in methylenetetrahydrofolate reductase and thymidylate synthase were significantly associated with increased risk of esophageal squamous cell carcinoma, gastric cardia carcinoma, and pancreatic carcinoma
Literature	GENERIF_SUMMARY	Ts mRNA is induced by preoperative chemoradiotherapy in both responders and nonresponders to combined modality treatment in rectal carcinoma.
Literature	GENERIF_SUMMARY	single nucleotide polymorphism of thymidylate synthase in colorectal cancer (Review)
Literature	GENERIF_SUMMARY	Expression of thymidylate synthase and thymidine phosphorylase mRNA is a useful predictive parameter for the survival of postoperative gastric cancer patients after 5-fluorouracil-based adjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Common variation in MTHFR and TYMS genes may be associated with RCC risk, particularly when vegetable intake is low.
Literature	GENERIF_SUMMARY	Examined association between MTHFR/thymidylate synthase gene polymorphisms and methylation of p16(INK4A) and hMLH1 genes in spontaneously aborted embryos with normal chromosomal integrity.
Literature	GENERIF_SUMMARY	There is a relationship between TS genotype and TS protein expression in clinical specimens of colon adenocarcinoma
Literature	GENERIF_SUMMARY	Genetic variation is not useful for predicting toxicity from capecitabine in patients with advanced colorectal cancer.
Literature	GENERIF_SUMMARY	findings do not suggest a significant association of MTHFR/TS allele/genotype with methotrexate response in our ethnically distinct Indian (Asian) rheumatoid arthritis patient
Literature	GENERIF_SUMMARY	This study reports the differential TS expression in the spectrum of neuroendocrine tumors and indicates TS as a possible marker of treatment efficacy in well-differentiated gastroenteropancreatic neuroendocrine carcinomas patients treated with 5-FU.
Literature	GENERIF_SUMMARY	increased expression of the TYMS oncogene may be of importance for the development of human intracranial meningiomas
Literature	GENERIF_SUMMARY	at least one-third of hTS populates the inactive conformer

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	demonstrates increased survival for neoadjuvant treated esophageal adenocarcinoma with TS, ERCC1, or GSTP-1 mRNA level at or below the median
Literature	GENERIF_SUMMARY	Thymidylate synthase genotypes were not associated with thrombosis and serum homocysteine concentration in Behcets patients; mean serum homocysteine level in patients with thrombosis was significantly higher than in patients without thrombosis
Literature	GENERIF_SUMMARY	Microsatellite instability and the association with plasma homocysteine and thymidylate synthase was determined in colorectal cancer.
Literature	GENERIF_SUMMARY	The genotype 6bp-/6bp- in the region untranslated 3' of TS gene, in the analyzed sample participates in important way in the development of CRC of the Mexican population.
Literature	GENERIF_SUMMARY	The transcriptional status of four key genes, thymidylate synthase (TYMS), MORF-related gene X (MRGX), Bcl2-antagonist/killer (BAK), and ATPase, Cu(2+) transporting beta polypeptide (ATP7B), can accurately predict response to 5-FU.
Literature	GENERIF_SUMMARY	prognostic factor for survival in colorectal carcinoma with radical resection and 5-FU chemotherapy; postoperative survival was significantly better in patients with low TS activity
Literature	GENERIF_SUMMARY	functional inactivity and mutations of p53 differentially affect TS, potentially influencing response to TS inhibitor-based treatment
Literature	GENERIF_SUMMARY	findings indicated that the TS gene polymorphism holds important prognostic information for stage II colon cancer; 5 year survival rate was significantly higher in patients with TS homozygocity than in heterozygous patients
Literature	GENERIF_SUMMARY	No association was detected between the Single Nucleotide Polymorphism in the Thymidylate synthase and susceptibility to hemifacial spasm.
Literature	GENERIF_SUMMARY	no association between either green tea intake or gene polymorphisms of MTHFR (C677T and A1298C) and TYMS (1494 ins/del) and breast cancer risk.
Literature	GENERIF_SUMMARY	Analysis of in vivo C-MYC interactions with TS, IMPDH2 and PRPS2 genes confirmed that they are direct C-MYC targets.
Literature	GENERIF_SUMMARY	The polymorphisms of TS 3'-UTR ins6/del6 are associated with gastric cancer patients treated with fluorouracil-based adjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Copy-number analysis of TYMS genes in frozen and FFPE DNAs of colorectal cancers is reported.
Literature	GENERIF_SUMMARY	The relationship of nuclear to cytoplasmic thymidylate synthase expression, given as a ratio of continuous AQUA scores, is a strong predictor of colon cancer

Main Category	Category	Terms
		survival.
Literature	GENERIF_SUMMARY	Bcl-2 expression significantly correlates with thymidylate synthase expression in colorectal cancer patients.
Literature	GENERIF_SUMMARY	polymorphisms in the 5'UTR and 3'UTR of thymidylate synthase may be associated with gastric cancer susceptibility
Literature	GENERIF_SUMMARY	Mutations of MTHFR and TSER are not likely significant risk factors of idiopathic recurrent spontaneous abortion in Korean women.
Literature	GENERIF_SUMMARY	MTHFR and TS gene variants are associated with decreased acute lymphoblastic leukemia risk
Literature	GENERIF_SUMMARY	Measurement of TS and ERCC1 mRNA levels in primary colorectal cancer can predict those in synchronous liver metastases, but not in metachronous ones.
Literature	GENERIF_SUMMARY	head & neck, gastric, colorectal, breast, lung & pancreatic cancer were examined; findings show mRNA expression & protein level of thymidylate synthase, dihydropyrimidine dehydrogenase & orotate phosphoribosyltransferase differed according to cancer type
Literature	GENERIF_SUMMARY	Combination of polymorphisms within 5' and 3' untranslated regions of thymidylate synthase gene modulates response to fluorouracil in colorectal cancer.
Literature	GENERIF_SUMMARY	No correlation was found between thymidylate synthase protein expression and gene polymorphism in patients with colorectal carcinoma
Literature	GENERIF_SUMMARY	Microsatellite instability was correlated with TS gene expression but not significantly with DPD.
Literature	GENERIF_SUMMARY	rs3819102, an htSNP located in the 3'-flanking region of the TYMS gene and an intron of the ENOSF1 gene, was found to be associated with an increased risk of endometrial cancer.
Literature	GENERIF_SUMMARY	recurrence-free survival time curve showed significant differences when stratified by tumor diameter, ER expression, and TS levels, but not by menopausal status, nodal status, surgical method, p53 expression, DPD activity, or HER2 expression
Literature	GENERIF_SUMMARY	Allelic imbalance of 18p11.32 and 18p11.31 at the TYMS locus was detected in renal cell carcinomas.
Literature	GENERIF_SUMMARY	The importance of the TYMS gene variants in response to 5-FU treatment.
Literature	GENERIF_SUMMARY	A polymorphism 381 A>G (E127E) in thymidylate synthase gene was successfully identified in children using RT-PCR-DGGE combined with DNA sequencing.
Literature	GENERIF_SUMMARY	None of the TYMS polymorphisms could explain 5FU toxicity or clinical outcome in colorectal cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Patients with a thymidylate synthase (TS)polymorphism expressed as 2R/3R might be more sensitive to 5-FU regimen than those with a polymorphism expressed as 3R/3R.
Literature	GENERIF_SUMMARY	These results demonstrate significant differences in nuclear TS expression between proximal and distal cancers and suggest the potential importance of the site of the tumor for proper stratification of patients for chemotherapy.
Literature	GENERIF_SUMMARY	Loss of heterozygosity at thymidylate synthase locus is a frequent and early event in Barrett's metaplasia, dysplasia, and carcinoma sequences sequence.
Literature	GENERIF_SUMMARY	Examine role of thymidylate synthase polymorphisms in patient response to gastric cancers treated with FOLFOX or FOLFIRI.
Literature	GENERIF_SUMMARY	thymidylate synthase/ribonucleotide reductase gene silencing and deoxycytidine kinase
Literature	GENERIF_SUMMARY	High TS in the cancerous tissue was associated with better disease-free survival in biliary tract cancers.
Literature	GENERIF_SUMMARY	Ectopic expression of wild-type CaSR in colon carcinoma cells inhibited the expression of TS. Vitamin D suppressed the expression of TS and TS gene transcriptional activities.
Literature	GENERIF_SUMMARY	Polymorphisms in the enhancer region of the thymidylate synthase gene are associated with thymidylate synthase levels in normal tissues but not in malignant tissues of patients with colorectal cancer.
Literature	GENERIF_SUMMARY	patients who developed distant recurrence of rectal cancer were found to have significantly higher intratumoral thymidylate synthase (TS), dihydropyrimidine dehydrogenase (DPD) and orotate phosphoribosyl transferase (OPRT)
Literature	GENERIF_SUMMARY	Data show that MTHFR and TYMS genotypes influence TPMT activity, and that males and females demonstrate differential modulation of TPMT activity.
Literature	GENERIF_SUMMARY	Poorly-differentiated cells showed increased expression of thymidylate synthase in patients with esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	No asssociation between TYMS polymorphisms and both, follicular lymphoma and diffuse, large B-Cell lymphoma was observed in a cohort of swedish patients.
Literature	GENERIF_SUMMARY	Genetic variations in thymidylate Synthase leads to hand-foot syndrome in metastatic colorectal treated with 5-fluorouracil based therapy.
Literature	GENERIF_SUMMARY	polymorphisms in TYMS, MTHFR, p53 and MDR1 may have roles in breast cancer
Literature	GENERIF_SUMMARY	Results suggested that intratumoral TS expression is an

Main Category	Category	Terms
		independent prognostic factor in patients with gastric cancer who received postoperative adjuvant chemotherapy with S-1.
Literature	GENERIF_SUMMARY	Polymorphisms of thymidylate synthase is associated with colorectal carcinoma.
Literature	GENERIF_SUMMARY	thymidylate synthase expression predicts disease free and overall survival in selected gastric cancer patients treated with 5-FU-based adjuvant chemotherapy after curative resection.
Literature	GENERIF_SUMMARY	protein expression neither influenced the prognosis nor predicted the response to pemetrexed in Korean patients with advanced non-small cell lung cancer
Literature	GENERIF_SUMMARY	Increased thymidylate synthase expression is associated with bladder cancers.
Literature	GENERIF_SUMMARY	The proportion of individuals who are homozygous for the 2R allele of the 5'UTR TYMS polymorphism is less in individuals who are homozygous for the T allele of MTHFR 677 than in individuals homozygous for the C allele of MTHFR 677.
Literature	GENERIF_SUMMARY	N-terminal extension affects the conformational state of the human thymidylate synthase catalytic region.
Literature	GENERIF_SUMMARY	Thymidylate synthase genotype may help to identify patients with stage II/III rectal cancer who could benefit from pre- and postoperative fluorouracil-based chemotherapy.
Literature	GENERIF_SUMMARY	no association was observed with respect to MTHFR C677T and A1298C, RFC A80G, and TYMS polymorphisms and risk of retinoblastoma
Literature	GENERIF_SUMMARY	Thymidine kinase-1 and thymidylate synthase expression was markedly different between cancer types, suggesting that response to TAS-102 may differ.
Literature	GENERIF_SUMMARY	The present study supports that the effect of TS polymorphisms on the clinical evolution of advanced colorectal cancer patients is significantly influenced by gender.
Literature	GENERIF_SUMMARY	carriers of the TS 3R/3R genotype were at increased risk of CIMP- colorectal cancer
Literature	GENERIF_SUMMARY	Data suggest that TS gene expression (mRNA) can be a useful marker of better survival in patients with stage-II colon cancers receiving 5-FU adjuvant chemotherapy, independently of microsatellite instability.
Literature	GENERIF_SUMMARY	Advanced lung adenocarcinoma patients with low TS expression had significantly higher control rate by pemetrexed treatment than those with TS overexpression.
Literature	GENERIF_SUMMARY	Significantly higher mRNA and protein levels were associated with colorectal adenocarcinomas.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Suggest that the position, rather than the number of functional USF E-box repeats in the thymidylate synthase enhancer region, is responsible for determining gene expression in vitro and TS activity in vivo.
Literature	GENERIF_SUMMARY	TYMS rs495139 may be associated with a differential risk of ovarian carcinoma types, indicating the importance of accurate histopathologic classification.
Literature	GENERIF_SUMMARY	The objective of the present study was to analyze the TYMS gene polymorphism in patients with Breast cancer, and evaluate its response to 5FU
Literature	GENERIF_SUMMARY	Influence of thymidylate synthase gene polymorphisms on total plasma homocysteine concentrations
Literature	GENERIF_SUMMARY	Observed allelic frequencies in breast cnacer were
Literature	GENERIF_SUMMARY	miR-192 and miR-215 target thymidylate synthase expression in colorectal cancer cell lines.
Literature	GENERIF_SUMMARY	2R/2R-6 bp/6 bp combined genotype of thymidylate synthase may be related to high gastric cancer susceptibility.
Literature	GENERIF_SUMMARY	TYMP and TYMS mRNA and protein expression levels were significantly higher in epithelial ovarian cancer, and DPYD mRNA and protein expression levels were significantly lower.
Literature	GENERIF_SUMMARY	Results provide information on the roles of specific subregions within the intrinsically disordered N-terminal domain of hTS in regulation of degradation underlying the ubiquitin-independent proteasomal degradation pathway.
Literature	GENERIF_SUMMARY	Variants in the dihydropyrimidine dehydrogenase, methylenetetrahydrofolate reductase and thymidylate synthase genes predict early toxicity of 5-fluorouracil in colorectal cancer patients.
Literature	GENERIF_SUMMARY	Thymidylate synthase, thymidine phosphorylase, and dihydropyrimidine dehydrogenase play a role in the prognosis of non-small-cell lung cancer.
Literature	GENERIF_SUMMARY	This study suggests that single nucleotide polymorphisms within the TS enhancer region affect the tumor response to preoperative 5-FU-based chemoradiotherapy in rectal cancer.
Literature	GENERIF_SUMMARY	Data show that the expression of TS provides an alternative source of substrate for DNA synthesis and positively correlates with GEM resistance and shortened patient survival.
Literature	GENERIF_SUMMARY	upregulated expression is a marker for predicting postoperative recurrence in patients with lung adenocarcinoma
Literature	GENERIF_SUMMARY	Breast tissue with the +6 bp/+6 bp genotype had a significantly higher TYMS protein expression than did

Main Category	Category	Terms
		those with other genotypes.
Literature	GENERIF_SUMMARY	in-situ TS1 protein expression in Ewing/primitive neuroectodermal tumor is a prognostic marker.
Literature	GENERIF_SUMMARY	Phosphorylation of human thymidylate synthase by casein kinase 2 is selective for the inactive conformation.
Literature	GENERIF_SUMMARY	Tumor expression levels of thymidylate synthetase and dihydropyrimidine dehydrogenase are predictive of response to S-1-carboplatin chemotherapy in patients with advanced non-small cell lung cancer
Literature	GENERIF_SUMMARY	These data suggest Neil1 may be a critical mediator of base excision repair of incorporated dUMP following thymidylate synthetase pathway inhibition.
Literature	GENERIF_SUMMARY	Germline thymidylate synthase and XRCC1 polymorphisms might be useful as predictive markers of rectal tumor response to neoadjuvant chemoradiotherapy with 5-fluorouracil.
Literature	GENERIF_SUMMARY	the TSER polymorphism may increase susceptibility to breast cancer in the Caucasian population and the TS3'-UTR polymorphism may be a genetic determinant for developing breast cancer in the Asian population(Meta-analysis)
Literature	GENERIF_SUMMARY	crystal structures of human TS (hTS) complexes with FdUMP and dUMP were obtained, indicating that this form should facilitate high-throughput analysis of hTS complexes with drug candidates
Literature	GENERIF_SUMMARY	thymidylate synthase dimer contains an unusual S-turn motif that also occurs in the ;off' state of the human ribosomal decoding site RNA
Literature	GENERIF_SUMMARY	Conotruncal and related defects (CTRDs) risk may be related to variation in folate pathways genes, NOS3 and TYMS through the maternal genotype.
Literature	GENERIF_SUMMARY	Data indicate there were no significant associations among ERCC1 protein expression, TS and ERCC1 polymorphisms, and clinical outcome.
Literature	GENERIF_SUMMARY	serum s-adenosylhomocysteine levels are significantly effected by a combined thymidylate synthetase 3'UTR, and methylenetetrahydrofolate reductase C677T polymorphisms
Literature	GENERIF_SUMMARY	Data show that polymorphisms in the thymidylate synthase gene are strongly associated with d4T-TP intracellular levels and with development of HALS.
Literature	GENERIF_SUMMARY	Lower TS expression in adenocarcinoma of the lung was confirmed in a large-scale study.
Literature	GENERIF_SUMMARY	Low Thymidylate synthase expression is associated with treatment response in advanced rectal cancer.
Literature	GENERIF_SUMMARY	Polymorphisms in the TS gene influenced its protein

Main Category	Category	Terms
		expression and affected sensitivity of 5-FU in seven pancreatic cancer cell lines.
Literature	GENERIF_SUMMARY	MTHFR and TYMS polymorphisms did not show significant association with survival in head and neck squamous cell carcinoma patients
Literature	GENERIF_SUMMARY	Data show there was no significant differences among the groups regarding XRCC1 codons 399 and 194 and ERCC2 codons 751 and 312 polymorphisms, but the TYMS 3R variant allele was significantly associated with a reduced risk of childhood ALL.
Literature	GENERIF_SUMMARY	Our results indicate that MTHFR activity and a specific combination of the MTHFR 1298A>C and TYMS 3'-UTR ins/del polymorphisms are possible predictors of 5-FU treatment-related toxicity.
Literature	GENERIF_SUMMARY	Promoter tandem repeat polymorphism of the thymidylate synthase gene is associated with relapse frequency in in childhood acute lymphoblastic leukemia
Literature	GENERIF_SUMMARY	The thymidylate synthase genotype and XRCC1 Arg399Gln polymorphism might help to identify Stage II-III rectal cancer patients with a better outcome after preoperative concomitant chemoradiotherapy.
Literature	GENERIF_SUMMARY	These data suggest that all polymorphisms coding for MTHFR, MTR, MTRR and TS have consistent roles in the increased risk of sporadic colorectal adenocarcinoma among the southeastern population of Brazil.
Literature	GENERIF_SUMMARY	There were no significant associations of the polymorphisms of TYMS 3R-->2R or MTR 2756 A-->G with the risk of NSCLC
Literature	GENERIF_SUMMARY	Data suggest that thymidylate synthase (TS) and Src may belong to a common pathway that bears prognostic significance in NSCLC, and that Src represents a potential target to improve the efficacy of TS-inhibiting agents.
Literature	GENERIF_SUMMARY	A thymidylate synthase 28-bp repeat polymorphisms is not a risk factor for susceptibility to acute lymphoblastic leukemia amongst Kurdish children in Western Iran.
Literature	GENERIF_SUMMARY	Low thymidylate synthase expression is associated with treatment response in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Our current study found that higher thymidylate synthetase (TS)levels are associated with an adverse response to bevacizumab therapy, indicating that TS is an effective biomarker for predicting response to bevacizumab therapy.
Literature	GENERIF_SUMMARY	Baseline TS and ERCC-1 levels may allow the selection of patients who benefit from FOLFOX4 chemotherapy
Literature	GENERIF_SUMMARY	Study used transient transfection, antibiotic-selected pools of transfected cells, and stably transfected clones to study thymidylate synthase 5'-untranslated region as

Main Category	Category	Terms
		a regulator of translation efficiency.
Literature	GENERIF_SUMMARY	Thirty (39.47%) patients were found to have higher TS expression in primary tumors with earlier stage (P = 0.018), lower histological grades (P = 0.001) and high frequency microsatellite instability (P = 0.000).
Literature	GENERIF_SUMMARY	Results provide support for the role of methylenetetrahydrofolate reductase and thymidylate synthase polymorphism in gliomagenesis.
Literature	GENERIF_SUMMARY	Specific combinations of functional polymorphisms in DPYD and TYMS were demonstrated to be associated with DFS and overall survival in patients receiving adjuvant 5-FU-based treatment
Literature	GENERIF_SUMMARY	Thymidylate synthase expression analyses associated homozygous three repeats (3R/3R) to higher TS expression
Literature	GENERIF_SUMMARY	A positive TS expression was a powerful prognostic factor to predict a poor outcome in completely resected non-small cell lung cancer patients
Literature	GENERIF_SUMMARY	no significant association was found between TYMS or the MTR polymorphisms and the risk of primary liver cancer
Literature	GENERIF_SUMMARY	This study investigated the expression of TS and DPD in cervical lymph node metastases and its relationship with primary oral squamous cell carcinoma, as well as the interaction between these enzymes and Kangai 1(KAI1/CD82).
Literature	GENERIF_SUMMARY	High TS and DPD mRNA levels on FFPE specimens may predict distant recurrence of rectal cancer treated with 5-FU-based preoperative CRT followed by surgery.
Literature	GENERIF_SUMMARY	An important role of polymorphisms and gene-gene interactions within the folate pathway in high dose methotrexate-related toxicity in childhood acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	High thymidylate synthase is associated with treatment response in colorectal cancer.
Literature	GENERIF_SUMMARY	TS mRNA expression did not correlate with any clinicopathological variables, including prognosis.
Literature	GENERIF_SUMMARY	Studies indicate that Thymidylate synthase (TYMS) 3'-untranslated region (TS3'-UTR) polymorphisms may have a protective effect in hematological malignancies.
Literature	GENERIF_SUMMARY	Data sugget that inhibition of dUTPase is a mechanism-based therapeutic approach to significantly enhance the efficacy of thymidylate synthase (TS)-targeted chemotherapeutic agents.
Literature	GENERIF_SUMMARY	Thymidylate synthase and p53 positivity was significantly associated with the failure of FEC treatment.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Thymidylate synthase gene expression may help identify patients at increased risk of the progression of gallbladder carcinoma.
Literature	GENERIF_SUMMARY	meta-analysis suggested thymidylate synthase enhancer region (TSER) and TS1494del6 polymorphisms are not associated with cancer risk when all studies were pooled; found an association between TSER polymorphism and cancer risk among Asians but not Caucasians; TS1494del6 polymorphism may contribute to susceptibility of breast cancer among Asians
Literature	GENERIF_SUMMARY	there may be no association between common genetic polymorphisms of the regulatory region of the TYMS gene and congenital cardiac septal defects in the Han Chinese population
Literature	GENERIF_SUMMARY	There is a small but statistically significant association between treatment effect (both intended effects and adverse events) and a TYMS genotype associated with low protein expression.
Literature	GENERIF_SUMMARY	A meta-analysis suggested that both 1053C > T and IVS6-68C >T TYMS polymorphisms were significantly associated with decreased risks of non-Hodgkin lymphoma among Caucasians.
Literature	GENERIF_SUMMARY	Japanese men who were homozygous for the thymidylate synthase double-repeat allele had a slightly decreased risk of colorectal adenomas as compared with those homozygous for the thymidylate syntahse triple-repeat allele
Literature	GENERIF_SUMMARY	Multifactorial pharmacogenetic analysis in colorectal cancer patients receiving 5-fluorouracil-based therapy
Literature	GENERIF_SUMMARY	Polymorphisms and haplotype structure of the human thymidylate synthase gene is associated with response to therapy in pancreatic cancer.
Literature	GENERIF_SUMMARY	High thymidylate synthase expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	TS PE and GCN vary widely in non-small cell lung and correlate significantly with each other. TS GCN is higher in SCCs, whereas TS PE does not associate with histological subtypes, clinical features, or survival.
Literature	GENERIF_SUMMARY	study concluded thatthe thymidylate synthase enhancer region (TSER) polymorphism appears not to be a risk factor for susceptibility to acute lymphoblastic leukemia in the Kashmir population
Literature	GENERIF_SUMMARY	thymidylate synthase(TS) protein expression tends to be related TS mRNA expression and is an independent prognostic factor in advanced colorectal cancer.
Literature	GENERIF_SUMMARY	mRNA expression of TYMS, DPYD, and TYMP is associated with distinct characteristics and may be useful for predicting survival in patients with stage IV colorectal cancer.

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Literature	GENERIF_SUMMARY	High expression levels of orotate phosphoribosyl transferase and thymidylate synthase in colorectal cancer appear to be significantly involved in metastasis after curative surgery
Literature	GENERIF_SUMMARY	study did not find an association between TS enhancer region or TS 1494del6 polymorphisms and premature ovarian failure
Literature	GENERIF_SUMMARY	The 3 G/3 G genotype of the TYMS gene may indicate predisposition of poor response to methotrexate in rheumatoid arthritis patients.
Literature	GENERIF_SUMMARY	Thymidylate synthase may be helpful as a prognostic marker in lymph node negative colorectal cancer.
Literature	GENERIF_SUMMARY	The variant del14946bp allele within the thymidylate synthase gene single nucelotide polymorphism were associated with the development of pancreatic cancer.
Literature	GENERIF_SUMMARY	Upon multivariate analysis, aberrant intratumoral TYMS in NSCLC predicted for shorter progression free survival than normal TYMS in 1st line chemo-naive treated patients.
Literature	GENERIF_SUMMARY	TS 5'UTR 28-bp tandem repeats and 3'UTR 6-bp ins/del polymorphisms may not be associated with cancer risk.
Literature	GENERIF_SUMMARY	observe a 3.6-fold increase in CTD risk among infants who were homozygotes for the 6 bp deletion in the 3'-untranslated region of thymidylate synthase and whose mothers had low folate intake during the peri-conceptional period
Literature	GENERIF_SUMMARY	High thymidylate synthase expression is associated with response to therapy in non-small-cell lung cancer.
Literature	GENERIF_SUMMARY	2R/3R polymorphisms in the thymidylate synthase 5'-untranslated enhanced region is associated with gastric cancer.
Literature	GENERIF_SUMMARY	meta-analysis suggests that the 3R variant of Thymidylate synthase 5'-untranslated enhanced region 2R/3R polymorphism contributes to gastric cancer risk in the Caucasian population, while any association in Asian populations needs further study
Literature	GENERIF_SUMMARY	Polymorphisms in the thymidylate synthase gene are associated with colorectal cancer.
Literature	GENERIF_SUMMARY	TYMS 1494ins/del, 1053C>T and IVS6-68C>T polymorphisms can be prognostic markers for OS in patients with EOC, independently from stage at diagnosis, median age and tumor histotype
Literature	GENERIF_SUMMARY	There is an association between the 2R/3R thymidylate synthase variants and the development of sporadic breast cancer, and 2R/2R variants and the development of hereditary breast cancer.
Literature	GENERIF_SUMMARY	Report essential roles for ribonucleotide reductase and thymidylate synthase in C-MYC-dependent suppression

Main Category	Category	Terms
		of senescence in melanoma cells.
Literature	GENERIF_SUMMARY	Thymidylate synthase 3'-UTR ins6/ins6 genotype could predict the poor survival of advanced gastric cancer patients treated with capecitabine plus paclitaxel
Literature	GENERIF_SUMMARY	POlymorphsim in TS shwoed no impact on risk of development of childhood acute leukemia.
Literature	GENERIF_SUMMARY	This study showed that TS protein is highly expressed in NSCLC and that polymorphisms of TS 3'-UTR 1494del 6 bp are associated with sensitivity of lung adenocarcinoma patients to pemetrexed treatment
Literature	GENERIF_SUMMARY	Low expression TS plus a MTHFR genotype associated with increased activity is associated with the development of peripheral neuropathy in d4T-exposed patients
Literature	GENERIF_SUMMARY	Thymidylate synthase expression does not correlate with prognosis of colorectal liver metastases.
Literature	GENERIF_SUMMARY	Low thymidylate synthase expression is significantly associated with better response rate and progression-free survival in patients with pulmonary adenocarcinoma who were treated with pemetrexed/ cisplatin as first-line chemotherapy.
Literature	GENERIF_SUMMARY	Thymidylate synthase gene copy number is associated with response to pemetrexed treatment of lung adenocarcinoma.
Literature	GENERIF_SUMMARY	Thymidylate synthase rs34743033 tandem repeat polymorphism is associated with stomatitis in childhood acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	Thymidylate synthase expression determines pemetrexed targets and resistance development in tumor cells.
Literature	GENERIF_SUMMARY	Data indicate the BGC 945 formed complexes and inhibited E coli and human thymidylate synthase (TS).
Literature	GENERIF_SUMMARY	High thymidylate synthase expression is associated with response to therapy in nonsquamous non-small-cell lung cancer.
Literature	GENERIF_SUMMARY	High TYMS immunoexpression correlated with primary nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	Variation in the DPYD, TYMS, CDA and MTHFR genes are clinically significant predictors of fluoropyrimidine toxicity.
Literature	GENERIF_SUMMARY	High TMYS expression is associated with non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Overexpression of thymidylate synthase correlates with lymphatic metastasis and might serve as a prognostic factor for the poor outcome in colorectal cancer patients.

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Literature	GENERIF_SUMMARY	Study of genetic risk of prevalent hrHPV infections in Nigerian women found significant associations with SNPs on ribosomal protein gene S19 (RPS19) and Thymidylate Synthase gene (TYMS), in an allelic model. This risk remained significant, after adjusting for age, body mass index, smoking, age at menarche, age at sexual debut, lifetime total number of sexual partners and the total number of pregnancies.
Literature	GENERIF_SUMMARY	Data suggest that methylenetetrahydrofolate reductase (NAD(P)H), dihydrofolate reductase, thymidylate synthetase and SLC19A1 genes present increased expression after the highest dose of methotrexate in laryngeal cancer cell line.
Literature	GENERIF_SUMMARY	berberine suppresses the growth of cDDP-resistant cells more than the sensitive counterparts, by interfering with the expression of folate cycle enzymes, dihydrofolate reductase (DHFR) and thymidylate synthase (TS).
Literature	GENERIF_SUMMARY	High TYMS expression levels are associated with resistance to 5-fluorouracil in cancer.
Literature	GENERIF_SUMMARY	early identification of individuals at risk of disease along with improvement in the prediction of patients' outcome will offer a powerful tool for the translation of TYMS polymorphisms into clinical practice and individualization of treatments
Literature	GENERIF_SUMMARY	Proximal colorectal carcinoma tumors are characterized by higher TS expression genotypes than distal tumors, and are at significantly greater risk of early recurrence during the first 20 months after surgery.
Literature	GENERIF_SUMMARY	Low/negative TYMS expression was significantly associated with higher response rate and longer survival of advanced non-small cell lung cancer patients receiving pemtrexed-based chemotherapy. (Meta-analysis)
Literature	GENERIF_SUMMARY	Data suggest that expression of TYMS can be used as biomarker of response to pemetrexed-containing chemotherapy in non-small-cell lung cancer patients; lower expression of TYMS is associated with better response. [META-ANALYSIS; REVIEW]
Literature	GENERIF_SUMMARY	TS and RFC1 polymorphisms are associated with colorectal cancer prognosis in Korean patients.
Literature	GENERIF_SUMMARY	TYMS expression was not associated with treatment outcome in colorectal patients treated with fluorouracil.
Literature	GENERIF_SUMMARY	Carriers of the rs16430 0 bp variant allele had an increased risk of breast cancer [adjusted odd ratio (OR) = 1.37, 95% confidence interval (CI)
Literature	GENERIF_SUMMARY	Data suggest that assessing expression of thymidylate synthase and in conjunction with p21(WAF1)/p53 of colon cancers may improve the selection of patients suitable for 5-fluorouracil (5FU)-based adjuvant

Main Category	Category	Terms
		chemotherapy.
Literature	GENERIF_SUMMARY	The strong association of increased TS expression with aggressive disease and early death supports the role of Thymidylate synthase as a cancer promoting agent in these tumors.
Literature	GENERIF_SUMMARY	5'tandem repeat polymorphism of the thymidylate synthase gene is associated with risk of colorectal cancer.
Literature	GENERIF_SUMMARY	We suggest that a chip including DPYD, TYMS, TYMP, TK1, and TK2 genes is a potential tool to predict response in LARC following fluoropyrimidine-based CCRT.
Literature	GENERIF_SUMMARY	High is thymidylate synthetase associated with low response to chemotherapy in non-small-cell lung cancer.
Literature	GENERIF_SUMMARY	Fetal SNPs with the lowest Bayesian false-discovery probability (rs2612101, rs2847607, rs2847326, rs2847324) were found within the TYMS gene of conotruncal heart defect cases.
Literature	GENERIF_SUMMARY	the TYMS polymorphisms may be useful tools to predict which advanced NSCLC patients could benefit more from platinum-based chemotherapy regimens.
Literature	GENERIF_SUMMARY	Genome-wide association studies identify 10 novel genetic loci as risk factors for methotrexate response in rheumatoid arthritis patients additionally to polymorphism in DHFR, FPGS and TYMS.
Literature	GENERIF_SUMMARY	Of 36 previously assessed polymorphisms, only four-TYMS 5'VNTR 2R/3R, TYMS 3'UTR 6bpins-del, DPYD 2846TA, and DPYD *2A-were formally associated with global G3+ toxicity in our analysis. Associations were only present in fluorouracil monotherapy regimens.
Literature	GENERIF_SUMMARY	the thymidylate synthase 5'-UTR 28bp tandem repeat reduces risk for acute lymphoblastic leukemia and influences disease progression in acute lymphoblastic leukemia
Literature	GENERIF_SUMMARY	The toxicity polymorphism in the TYMS region may actually act through ENOSF1
Literature	GENERIF_SUMMARY	Overall, no significant association between the TS polymorphisms and CRC risk was found
Literature	GENERIF_SUMMARY	These results suggest that three genetic polymorphisms in the miRNA binding sites of TYMS may be associated with cancer risk and survival of gastric cancer patients.
Literature	GENERIF_SUMMARY	Colorectal tumors, deficient in mismatch repair, overexpress thymidylate synthetase and dihydropyrimidine dehydrogenase.
Literature	GENERIF_SUMMARY	combined analysis of TS VNTR and MTHFR 677C>T SNP revealed shortening of progression-free survival in

Main Category	Category	Terms
		synchronous carriers of polymorphic tandem repeat sequence allele in TS and two C alleles in MTHFR
Literature	GENERIF_SUMMARY	Low Thymidylate Synthase expression is associated with recurrence after resection of colorectal liver metastases.
Literature	GENERIF_SUMMARY	The expression of thymidylate synthase and its correlation with survival in stage II/III gastric cancer patients
Literature	GENERIF_SUMMARY	TS expression is associated with chemotherapy response, PFS, and OS in advanced GC patients treated with capecitabine alone chemotherapy.
Literature	GENERIF_SUMMARY	genetic polymorphism is associated with differences in disease control rate and progression-free survival in malignant pleural mesothelioma patients
Literature	GENERIF_SUMMARY	Our study reveals that TYMS polymorphisms could be important to help predicting clinical response to MTX in RA patients.
Literature	GENERIF_SUMMARY	Genetic association study of MTHFR C677T polymorphism and TYMS tandem repeats in oral squamous cell carcinoma (OSCC) risk showed only TYMS tandem repeat may be associated with OSCC development.
Literature	GENERIF_SUMMARY	Data show that the thymidylate synthase (TS) 5'-untranslated enhanced region (TSER) and the 6-bp insertion/deletion in the 3'-untranslated region (3'-UTR) single-nucleotide polymorphisms are not associated with breast cancer risk in Mexican patients.
Literature	GENERIF_SUMMARY	Data indicate that thymidylate synthase (hTS) enzyme recognizes the apical hairpin of the TS site 1 mRNA motif.
Literature	GENERIF_SUMMARY	Data indicate that miR-203 enhances 5-FU chemosensitivity via the downregulation of TYMS in colorectal cancer and suggests that miR-203 has the potential as a therapeutic strategy for 5-FU-resistant colorectal cancer.
Literature	GENERIF_SUMMARY	There were no other associations between single - nucleotide polymorphisms and the efficacy of MTX treatment. CONCLUSIONS
Literature	GENERIF_SUMMARY	TS gene polymorphisms have no influence on the incidence of high-dose methotrexate-related toxicities in childhood ALL
Literature	GENERIF_SUMMARY	Germline polymorphism of TYMS have a significant effect on toxicity and clinical outcome in patients receiving capecitabine-based chemotherapy for metastatic gastrointestinal malignancies.
Literature	GENERIF_SUMMARY	Homocysteine levels might effect the disease activity and toxicity of MTX but 2R and 3R polymorphisms in the Thymidylate synthase gene, were not related with methotrexate-related toxicity in rheumatoid arthritis

Main Category	Category	Terms
		patients receiving folate supplementation.
Literature	GENERIF_SUMMARY	Patients who had CTC count above the median (2 CTCs/mL) showed more TYMS expression.
Literature	GENERIF_SUMMARY	our data suggest that SNPs rs2847153 in TYMS gene may be a potential biomarker for predicting clinical outcome and personalized treatment in NSCLC patients.
Literature	GENERIF_SUMMARY	Therefore, TS genotypes, based on a G>C SNP in the TR sequence of the TS gene, may be a useful biomarker for predicting outcomes for fluoropyrimidine-based chemotherapy in patients with locally advanced/ metastatic small bowel adenocarcinoma
Literature	GENERIF_SUMMARY	Colorectal cancer patients with low expression of ERCC1 and thymidylate synthase have better response to therapy and show longer survival.
Literature	GENERIF_SUMMARY	SAHA and 5-FU act synergistically to inhibit cell growth and tumorigenicity in Hepatocellular carcinoma via the induction of cell-cycle arrest and apoptosis through a mechanism involving the inhibition of thymidylate synthase
Literature	GENERIF_SUMMARY	TS expression level was reduced more in dMMR cells after irinotecan treatment (p
Literature	GENERIF_SUMMARY	The study suggests that patients carrying rs183205964, a G>C substitution in the promoter enhancer region of TYMS, are at strongly increased risk of severe, potentially life-threatening, toxicity when treated with fluoropyrimidines.
Literature	GENERIF_SUMMARY	TYMS gene amplification predicts outcome of patients receiving pemetrexed with advanced non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Dihydrofolate Reductase and Thymidylate Synthase Transgenes Resistant to Methotrexate Interact to Permit Novel Transgene Regulation.
Literature	GENERIF_SUMMARY	Study suggests a novel function of HSP90-Src pathway in regulation of TYMS expression and acquisition of 5-FU resistance in colon cancer.
Literature	GENERIF_SUMMARY	Studies suggest that dietary folate intake and gender may modify the associations between 5,10-methylenetetrahydrofolate reductase (MTHFR) and thymidylate synthase (TS) polymorphisms and adult leukemia risk.
Literature	GENERIF_SUMMARY	TYMS overexpression was detected in 61 % of non-small cell lung cancer patients and low expression in 39 %.
Literature	GENERIF_SUMMARY	High expression level of TS might be negative prognostic factor for resected NSCLC patients.
Literature	GENERIF_SUMMARY	treatment response, survival, and the associations between KRAS mutation status and tumour expression

Main Category	Category	Terms
		levels of BRCA1, TYMS and SRC retrospectively in a cohort of patients with non-small cell lung cancer, were evaluated.
Literature	GENERIF_SUMMARY	Thymidylate Synthase Polymorphisms are associated with the Risk of Lung Cancer.
Literature	GENERIF_SUMMARY	High expression of TYMS is associated with Colorectal cancer.
Literature	GENERIF_SUMMARY	Our findings emphasize the pivotal relevance of RNAi as an effective tool for increasing the therapeutic efficacy of PMX, a cornerstone in the treatment regimens of Malignant mesothelioma thereby, raising the possibility for the development of a novel therapeutic strategy, combination therapy of TS-shRNA and PMX, that can surpass many of the currently applied, but less effective, therapeutic regimens against lethal MPM.
Literature	GENERIF_SUMMARY	This study identified a novel single repeat in the TYMS gene which might have an impact on the expression of this gene, which needs to be confirmed by functional studies.
Literature	GENERIF_SUMMARY	High expression level of TYMS is associated with neuroendocrine lung tumors.
Literature	GENERIF_SUMMARY	High titers of both TYMS and PDLIM1 serum autoantibodies were significantly more prevalent in breast cancer cases than in controls.
Literature	GENERIF_SUMMARY	Dihydrofolate reductase and thymidylate synthase form a complex in vitro and co-localize in normal and cancer cells.
Literature	GENERIF_SUMMARY	PCB followed by PB was effective and safe in our study group. This regimen was associated with acceptable toxicity and with prolonged OS. In our cohort both Ki67 and TS expression were important prognostic biomarkers.
Literature	GENERIF_SUMMARY	results indicate significant correlation between the Lauren and Goseki histopathological classifications of gastric cancer and TS expression in tumor cells, and suggest that high TS expression may be a positive prognostic marker with regard to DFS in patients with gastric cancer without involvement of regional lymph nodes who underwent radical surgical treatment and were not treated with preoperative chemotherapy.
Literature	GENERIF_SUMMARY	A PCR technique was used for genotyping TYMS-TSER.
Literature	GENERIF_SUMMARY	Data suggest the potential association of methylenetetrahydrofolate reductase (MTHFR), thymidylate synthase (TS) and TS enhancer region (TSER) polymorphisms with recurrent implantation failure (RIF) risk in Korean patients.
Literature	GENERIF_SUMMARY	These findings suggested that the down-regulation of AKT-mediated TS expression by salinomycin enhanced the cisplatin-induced cytotoxicity in nonsmall cell lung cancer cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	This study finds that measurement of tumor levels of thymidylate synthase is not helpful in assigning specific adjuvant treatment for colorectal cancer. It also highlights the importance of using prospective analyses within treatment clinical trials as the optimal method of determining biomarker utility.
Literature	GENERIF_SUMMARY	Metronomic S-1 dosing combined with TS silencing might represent an emerging therapeutic strategy for the treatment of patients with advanced colorectal carcinoma.
Literature	GENERIF_SUMMARY	polymorphisms of TS 5'-UTR 2R (double repeats)/3R (triple repeats) of a 28-bp sequence (11 articles) and 3'-UTR del6/ins6 (seven articles) were not significantly associated with increased risk of gastric cancer.
Literature	GENERIF_SUMMARY	In pancreatic neuroendocrine neoplasms, biochemical response (p = 0.005) and high dihydropyrimidine-dehydrogenase expression (p = 0.018) were predictive markers of response to 5-FU-based chemotherapy. Thymidylate-Synthase deficiency was an independent risk factor for shorter progression-free survival.
Literature	GENERIF_SUMMARY	Two SNPs (rs523230 and rs9967368) in TYMS gene were significantly associated with the overall survival of HCC patients.
Literature	GENERIF_SUMMARY	The expressions of TYMS, TUBB3 and STMN1 were significantly associated with the clinicopathological characteristics of age, gender and family history of gastric cancer, but not with differentiation, growth patterns, metastasis and TNM staging in patients with gastric cancer.
Literature	GENERIF_SUMMARY	Moreover, we found that genotypes rs34743033 3R/2R, rs16430 ins6/del6, rs1045642 CC or CT, and rs2032582 GG were beneficial predictors of clinical treatment outcome in AGC patients, suggesting some clinical implications in chemotherapy of a Chinese population.
Literature	GENERIF_SUMMARY	High thymidylate synthetase expression is associated with chemoresistance in glioblastoma multiforme.
Literature	GENERIF_SUMMARY	this meta-analysis suggested that thymidylate synthase enhancer region (TSER) variation in the TYMS gene was not related to susceptibility to develop pediatric acute lymphoblastic leukemia
Literature	GENERIF_SUMMARY	The Thymidylate synthase (TS)-ZEB1 association was confirmed in clinical specimens from lung tumours and in a genetic mouse model of pancreatic cancer with ZEB1 deletion. Interestingly, TS itself appeared to have a regulatory role in epithelial-to-mesenchymal in cancer cells, as TS knockdown could directly reduce the EMT phenotype, the migratory ability of cells, the expression of stem-like markers, and chemoresistance.
Literature	GENERIF_SUMMARY	High TOP2A expression was significantly associated with longer time to progression after EDP-M. TOP2A and TS proteins assessed by immunohistochemistry significantly correlated with mRNA expression.

Main Category	Category	Terms
		Immunohistochemical TOP2A expression was associated with a non-significant better response and longer TTP after EDP-M.
Literature	GENERIF_SUMMARY	Combined expression analyses of hENT1, TS, and DPD may predict long-term outcomes in patients with borderline resectable pancreatic cancer after neoadjuvant chemoradiotherapy
Literature	GENERIF_SUMMARY	Crystal structure of the active form of native human thymidylate synthase in the absence of bound substrates has been reported.
Literature	GENERIF_SUMMARY	The study provides crucial information regarding the importance of TYMS6bpdel/del genotype and associated hyperhomocysteinemia in susceptibility to preterm delivery, fetal death and low birth weight.
Literature	GENERIF_SUMMARY	Data suggest that TYMS exhibits a ligand-binding site in dimer interface, suggesting that cavity in dimer interface serves as an allosteric site to regulate conformational switching between active and inactive states of the enzyme.
Literature	GENERIF_SUMMARY	This study aimed to explore the associations between MTHFR or TS genetic polymorphisms and susceptibility to acute lymphocytic leukemia (ALL) in children.
Literature	GENERIF_SUMMARY	This study demonstrates a significant correlation between low TS expression levels and long-term prognosis of patients with lung cancer.
Literature	GENERIF_SUMMARY	positive expression of TS was a significant predictor of worse progression free survival in patients with clinical stage 1 cutaneous angiosarcoma
Literature	GENERIF_SUMMARY	High TS expression is associated with treatment response in mesothelioma.
Literature	GENERIF_SUMMARY	In MAGIC, patients with a TS 2R/2R genotype appeared to derive a larger benefit from perioperative ECF chemotherapy than patients with 3R containing genotypes.
Literature	GENERIF_SUMMARY	Study shows that the TYMS TSER 3R allele increases the risk of thymic lymphoid hyperplasia in AChR+ Myasthenia Gravis (MG) patients and that the 3R allele in the promoter enhancer region results in increased protein production required for the synthesis of DNA precursors.
Literature	GENERIF_SUMMARY	There was significantly higher thymidylate synthase (TS) expression in non-small cell lung cancer (NSCLC) tumor tissue comparing to normal lung tissue.
Literature	GENERIF_SUMMARY	the present results indicate that the EGFR mutation status and TS and ERCC1 expression can be used as the predictors of overall survival after subsequent second-line treatments for adenocarcinoma non-small-cell lung cancer
Literature	GENERIF_SUMMARY	A statistically significant association between

Main Category	Category	Terms
		methylenetetrahydrofolate reductase (MTHFR) 677T/T, thymidylate synthase (TS) 5'-untranslated region (UTR) 3RG, TS 3'-UTR -6 bp/-6 bp, XRCC1 399G/G genotypes and short survival was found in patients receiving best supportive care (BSC)by multivariate analysis
Literature	GENERIF_SUMMARY	thymidylate synthase appears to be a clinically relevant predictive biomarker in patients with resected colorectal liver metastases receiving systemic 5-FU
Literature	GENERIF_SUMMARY	Thymidylate synthase (TS) expression was positively correlated with lymphatic metastasis and negatively correlated with the overall survival (OS) in pancreatic ductal adenocarcinoma (PDA) patients.
Literature	GENERIF_SUMMARY	The results indicate that TS 3 -UTR 6-bp insertion and MTHFR 677T and 1298C alleles increase 5-flourouracil resistance in colorectal cancer cells.
Literature	GENERIF_SUMMARY	This meta-analysis demonstrates that the TYMS 2R/3R and 6bp I/D polymorphisms may not be associated with non-responsiveness to or toxicity of MTX therapy in RA patients [meta-analysis]
Literature	GENERIF_SUMMARY	results indicate XRCC3 Thr241Met and TYMS 5'-UTR VNTR polymorphisms are associated with time-to-metastasis, and may have potential biological roles in expediting the metastatic process
Literature	GENERIF_SUMMARY	The significant ethnic differences in the distributions of the TYMS rs2790 A > G polymorphism.
Literature	GENERIF_SUMMARY	TYMS SNPs are associated with susceptibility to osteoporosis and osteoporotic vertebral compression fractures in postmenopausal women.
Literature	GENERIF_SUMMARY	human (hs) TSase and its contrasts and the similarities to the well-studied Escherichia coli (ec) TSase
Literature	GENERIF_SUMMARY	TYMS was determined to form an octamer, depending on the presence of Cys43-disulfide.
Literature	GENERIF_SUMMARY	Deciphering Clinicoradiologic Phenotype for Thymidylate Synthase Expression Status in Patients with Advanced Lung Adenocarcinoma Using a Radiomics Approach.
Literature	GENERIF_SUMMARY	TYMS 2R3R polymorphism and the -[IVS]14+1G>A mutation in DPYD was not associated with susceptibility to CRC. However, the 2R/2R and 2R/3R genotypes of TYMS polymorphism could significantly contribute to hematological and gastric toxicity in CRC patients in this sample population
Literature	GENERIF_SUMMARY	No significant association has been found between rs495139 in the TYMS-ENOSF1 region and risk of ovarian carcinoma of mucinous histology. (Meta-analysis)
Literature	GENERIF_SUMMARY	24 hr. pretreatment with dexamethasone, but not same day co-administration of dexamethasone with therapy, impairs the TS inhibition-mediated ""flare"" in thymidine

Main Category	Category	Terms
		salvage pathway activity in non-small cell lung cancer."
Literature	GENERIF_SUMMARY	Polymorphism rs3819102 in thymidylate synthase gene is associated with lung cancer.
Literature	GENERIF_SUMMARY	The TYMS 6-bp deletion and 2R3R polymorphism were not associated with MTX-induced oral mucositis. Validation studies in prospective cohorts are necessary to assess the possible role of the low-expression TYMS genotypes in relation to MTX-induced oral mucositis.
Literature	GENERIF_SUMMARY	Results showed that the dominant alleles in three genes (DHFR, MTHFR, TYMS) were associated with hematologic toxicity, but none of them was associated with methotrexate level, hepatotoxicity or renal function
Literature	GENERIF_SUMMARY	FOXM1 modulates 5-FU resistance in colorectal cancer through regulating TYMS expression.
Literature	GENERIF_SUMMARY	Thymidylate synthase maintains the de-differentiated state of triple negative breast cancers.
Literature	GENERIF_SUMMARY	High expression of Thymidylate Synthase is associated with resistance of 5-Fluorouracil in Oral Squamous Cell Carcinoma
Literature	GENERIF_SUMMARY	Thymidylate synthetase enhancer region polymorphisms were associated with susceptibility to metabolic syndrome in Korean population.
Literature	GENERIF_SUMMARY	The immunoreactive thymidylate synthase (TS) signal in the cytoplasm of myoepithelial cells (MECs) was variable in 22 out of 32 cases of invasive breast carcinoma and 4 out of 20 cases of ductal carcinoma in situ. No immunoreactive TS signal was found in the nuclei of luminal and stromal cells in breast lesions, TS is a sensitive and specific MEC biomarker in the differential diagnosis of breast lesions.
Literature	GENERIF_SUMMARY	TYMS was absent in 100% of circulating tumor cells (CTCs) from locally advanced rectal cancer patients with pathologic complete response yet was expressed in 83% of non-responders prior to surgery (S2). RAD23B was expressed in CTCs from 75% of non-responders prior to neoadjuvant chemoradiation (S1) and in 100% of non-responders at S2. 100% of non-responders expressed TYMS mRNA at both timepoints.
Literature	GENERIF_SUMMARY	High TS expression predicts poor outcome after resection of hepatocellular carcinoma. Molecular markers might be robust predictors of patient outcome after resection of hepatocellular carcinoma
Literature	GENERIF_SUMMARY	Positive Cooperativity in Substrate Binding by Human Thymidylate Synthase.
Literature	GENERIF_SUMMARY	Our study unveils a mechanism of how TYMS is transcriptionally regulated by MYC, and provides rationales for the precise use of TYMS inhibitors in the clinic.
Literature	GENERIF_SUMMARY	Variants in TYMS, SLC19A1 and DHFR genes are

Main Category	Category	Terms
		potential biomarkers of myelotoxicity and could be used for 6-MP therapy individualization in maintenance phase of childhood ALL treatment, alongside with well-established TPMT variants
Literature	GENERIF_SUMMARY	Blood mRNA expression of REV3L and TYMS as potential predictive biomarkers from platinum-based chemotherapy plus pemetrexed in non-small cell lung cancer patients.
Literature	GENERIF_SUMMARY	Evaluating the role of ENOSF1 and TYMS variants as predictors in fluoropyrimidine-related toxicities
Literature	GENERIF_SUMMARY	upregulation in pancreatic cancer leads to unfavorable overall survival and recurrence-free survival
Literature	GENERIF_SUMMARY	MTHFR, TYMS and SLCO1B1 polymorphisms and adverse liver effects of methotrexate in rheumatoid arthritis.
Literature	GENERIF_SUMMARY	A Case-Control Study on the Correlation Between Thymidylate Synthase Gene Polymorphisms and Raltitrexed Treatment Combined with Transcatheter Arterial Chemoembolization in Hepatocellular Carcinoma Treatment.
Literature	GENERIF_SUMMARY	3'UTR polymorphism of Thymidylate Synthase gene increased the risk of persistence of pre-neoplastic cervical lesions.
Literature	GENERIF_SUMMARY	Identification of TYMS as a promoting factor of retroperitoneal liposarcoma progression
Literature	GENERIF_SUMMARY	Thymidylate synthase drives the phenotypes of epithelial-to-mesenchymal transition in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Effects of thymidylate synthase polymorphisms on toxicities associated with high-dose methotrexate in childhood acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	Polymorphisms in TYMS for Prediction of Capecitabine-Induced Hand-Foot Syndrome in Chinese Patients with Colorectal Cancer.
Literature	GENERIF_SUMMARY	Gene Polymorphisms Involved in Folate Metabolism and DNA Methylation with the Risk of Head and Neck Cancer.
Literature	GENERIF_SUMMARY	The prevalence and clinical relevance of 2R/2R TYMS genotype in patients with gastrointestinal malignancies treated with fluoropyrimidine-based chemotherapy regimens.
Literature	GENERIF_SUMMARY	Molecular Mechanisms and Tumor Biological Aspects of 5-Fluorouracil Resistance in HCT116 Human Colorectal Cancer Cells.
Literature	GENERIF_SUMMARY	Coptidis Rhizoma Extract Reverses 5-Fluorouracil Resistance in HCT116 Human Colorectal Cancer Cells via Modulation of Thymidylate Synthase.

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Literature	GENERIF_SUMMARY	Association between 5'-UTR and 3'-UTR Polymorphisms of the TYMS Gene and Breast Cancer in Kurdish Women of Iraq.
Literature	GENERIF_SUMMARY	Gender-dependent association of TYMS-TSER polymorphism with 5-fluorouracil or capecitabine-based chemotherapy toxicity.
Literature	GENERIF_SUMMARY	Thymidylate synthase is essential for efficient HIV-1 replication in macrophages.
Literature	GENERIF_SUMMARY	Association of MTHFR and TYMS gene polymorphisms with the susceptibility to HCC in Egyptian HCV cirrhotic patients.
Literature	GENERIF_SUMMARY	Significance of microRNA-330-5p/TYMS Expression Axis in the Pathogenesis of Colorectal Tumorigenesis.
Literature	GENERIF_SUMMARY	MicroRNA-140-3p represses the proliferation, migration, invasion and angiogenesis of lung adenocarcinoma cells via targeting TYMS (thymidylate synthetase).
Literature	GENERIF_SUMMARY	Investigation of DPYD, MTHFR and TYMS polymorphisms on 5-fluorouracil related toxicities in colorectal cancer.
Literature	GENERIF_SUMMARY	Germline thymidylate synthase deficiency impacts nucleotide metabolism and causes dyskeratosis congenita.
Literature	GENERIF_SUMMARY	DPYD and TYMS polymorphisms as predictors of 5 fluorouracil toxicity in colorectal cancer patients.
Literature	GENERIF_SUMMARY	Evaluation of TS and ENOSF1 Variants as a Biomarker in Response to Neoadjuvant Chemotherapy based on 5FU in Gastric Cancer Patients.
Literature	GENERIF_SUMMARY	Analysis of the TSER and G>C variants in the TYMS gene
Literature	GENERIF_SUMMARY	TYMS promotes genomic instability and tumor progression in Ink4a/Arf null background.
Literature	GENERIF_SUMMARY	Association of Thymidylate Synthase (TS) Gene Polymorphisms with Incidence and Prognosis of Coronary Artery Disease.
Literature	GENERIF_SUMMARY	LncRNA BC200 promotes the development of EBV-associated nasopharyngeal carcinoma by competitively binding to miR-6834-5p to upregulate TYMS expression.
Pathways	KEGG_PATHWAY	Pyrimidine metabolism
Pathways	KEGG_PATHWAY	One carbon pool by folate
Pathways	KEGG_PATHWAY	Metabolic pathways
Pathways	KEGG_PATHWAY	Nucleotide metabolism
Pathways	KEGG_PATHWAY	Antifolate resistance
Pathways	KEGG_PATHWAY	Folate transport and metabolism

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Pathways	REACTOME_PATHWAY	Metabolism
Pathways	REACTOME_PATHWAY	Metabolism of nucleotides
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Mitotic G1 phase and G1/S transition
Pathways	REACTOME_PATHWAY	Interconversion of nucleotide di- and triphosphates
Pathways	REACTOME_PATHWAY	G1/S-Specific Transcription
Pathways	REACTOME_PATHWAY	G1/S Transition
Pathways	REACTOME_PATHWAY	Cell Cycle, Mitotic
Pathways	WIKIPATHWAYS	Fluoropyrimidine activity
Pathways	WIKIPATHWAYS	miR targeted genes in leukocytes
Pathways	WIKIPATHWAYS	miR targeted genes in lymphocytes
Pathways	WIKIPATHWAYS	miR targeted genes in muscle cell
Pathways	WIKIPATHWAYS	One carbon metabolism
Pathways	WIKIPATHWAYS	Retinoblastoma gene in cancer
Pathways	WIKIPATHWAYS	Trans sulfuration one carbon metabolism and related pathways
Pathways	WIKIPATHWAYS	Circadian rhythm genes
Pathways	WIKIPATHWAYS	Ethanol effects on histone modifications
Pathways	WIKIPATHWAYS	Pyrimidine metabolism
Pathways	WIKIPATHWAYS	Pyrimidine metabolism and related diseases
Pathways	WIKIPATHWAYS	Disorders of folate metabolism and transport
Pathways	WIKIPATHWAYS	Biomarkers for pyrimidine metabolism disorders
Pathways	WIKIPATHWAYS	Serine metabolism
Pathways	WIKIPATHWAYS	ADHD and autism ASD pathways
Pathways	WIKIPATHWAYS	Pathways into methionine and folate cycles
Protein Domains	INTERPRO	Thymidylate_synthase
Protein Domains	INTERPRO	Thymidylate_synthase_AS
Protein Domains	INTERPRO	Thymidate_synth/dCMP_Mease_dom
Protein Domains	INTERPRO	Thymidate_synth/dCMP_Mease_sf
Protein Domains	INTERPRO	Thymidate_synth/dCMP_Mease